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MONOCHLOROTRIFLUOROPROPENE COMPOUNDS AND COMPOSITIONS AND METHODS USING SAME

Abstract

Various uses of monochlorotrifluoropropenes, in combination with one or more other components, including other fluoroalkenes, hydrocarbons; hydrofluorocarbons (HFCs), ethers, alcohols, aldehydes, ketones, methyl formate, formic acid, water, trans-1,2-dichloroethylene, carbon dioxide and combinations of any two or more of these, in a variety of applications, including as blowing agents, are disclosed.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. patent application Ser. No. 17/560,183, filed Dec. 22, 2021 (now pending), which is a divisional of U.S. patent application Ser. No. 17/071,601, filed Oct. 15, 2020 (now abandoned) which is a continuation of U.S. patent application Ser. No. 15/973,232, filed May 7, 2019 (now U.S. Pat. No. 10,828,579, issued Nov. 10, 2020) which application is a continuation of U.S. application Ser. No. 14/630,166, filed Apr. 24, 2015 (now abandoned), which is a continuation of U.S. application Ser. No. 13/394,289, filed Mar. 5, 2012, (now U.S. Pat. No. 8,962,707, issued Feb. 24, 2015), which is a national stage of Application No. PCT/US10/48036, filed Sep. 8, 2010, which claims priority to U.S. Provisional Patent Application No. 61/240,786, filed on Sep. 9, 2009. [0002] For the purposes of national phase entry into the U.S. only, the following additional priority claims are made. U.S. application Ser. No. 14/630,166 also claims the priority benefit as a continuation-in-part of U.S. application Ser. No. 12/351,807 filed Jan. 10, 2009 (now U.S. Pat. No. 9,499,729, issued Nov. 22, 2016), which is incorporated herein by reference as if fully set forth herein. The present application also claims the benefit of as a continuation-in-part, and incorporates by reference, each of the following United States Patent Applications: U.S. application Ser. No. 10/694,273, filed Oct. 27, 2003 (now U.S. Pat. No. 7,534,366); Ser. No. 11/385,259, filed Mar. 20, 2006, now pending, which in turn is a continuation of Ser. No. 10/695,212, filed Oct. 27, 2003, now abandoned; Ser. No. 10/694,272 filed Oct. 27, 2003 (now U.S. Pat. No. 7,230,146); Ser. No. 10/847,192, filed Aug. 29, 2007 (now U.S. Pat. No. 7,046,871), which in turn is a divisional of Ser. No. 10/837,525, filed Apr. 29, 2004 (now U.S. Pat. No. 7,279,451); Ser. No. 11/475,605, filed Jun. 26, 2006, now pending; and Ser. No. 12/276,137, filed Nov. 21, 2008, now pending, which claims the benefit of U.S. Provisional Application No. 60/989,997 filed Nov. 25, 2007 and U.S. application Ser. No. 11/474,887 filed Jun. 26, 2006, now pending, and PCT Application No. PCT/US07/64570, filed Mar. 21, 2007.

FIELD OF THE INVENTION

[0003] This invention relates to compositions, methods and systems having utility in numerous applications, including particularly heat transfer systems, such as refrigeration systems, blowing agents, foamable compositions, foams and articles made with or from foams, solvents, aerosols, propellants and cleaning composition. In preferred aspects, the present invention is directed to such compositions which comprise at least one monochlorotrifluoropropene.

BACKGROUND OF THE INVENTION

[0004] Fluorocarbon based fluids have found widespread use in many commercial and industrial applications, including as the working fluid in systems such as air conditioning, heat pump and refrigeration systems, as aerosol propellants, as blowing agents, as heat transfer media, and as

gaseous dielectrics. Because of certain suspected environmental problems, including the relatively high global warming potentials, associated with the use of some of the compositions that have heretofore been used in these applications, it has become increasingly desirable to use fluids having low or even zero ozone depletion potential, such as hydrofluorocarbons ("HFCs"). Thus, the use of fluids that do not contain chlorofluorocarbons ("CFCs") or hydrochlorofluorocarbons ("HCFCs") is desirable. Furthermore, some HFC fluids may have relatively high global warming potentials associated therewith, and it is desirable to use hydrofluorocarbon or other fluorinated fluids having as low global warming potentials as possible while maintaining the desired performance in use properties. Additionally, the use of single component fluids or azeotrope-like mixtures, which do not substantially fractionate on boiling and evaporation, is desirable in certain circumstances.

[0005] Certain fluorocarbons have been a preferred component in many heat exchange fluids, such as refrigerants, for many years in many applications. For, example, fluoroalkanes, such as chlorofluoromethane and chlorofluoroethane derivatives, have gained widespread use as refrigerants in applications including air conditioning and heat pump applications owing to their unique combination of chemical and physical properties. Many of the refrigerants commonly utilized in vapor compression systems are either single components fluids or azeotropic mixtures.

[0006] As suggested above, concern has been increasing in recent years about potential damage to the earth's atmosphere and climate, and certain chlorine-based compounds have been identified as particularly problematic in this regard. The use of chlorine-containing compositions (such as chlorofluorocarbons (CFC's), hydrochlorofluorocarbons (HCF's) and the like) as the working fluid in heat transfer systems, such as in refrigeration and air-conditioning systems, has become disfavored because of the ozone-depleting properties associated with many of such compounds. There has thus been an increasing need for new fluorocarbon and hydrofluorocarbon compounds and compositions that are attractive alternatives to the compositions heretofore used in these and other applications. For example, it has become desirable to retrofit chlorine-containing systems, such as blowing agent systems or refrigeration systems by replacing chlorine-containing compounds with non-chlorine-containing compounds that will not deplete the ozone layer, such as hydrofluorocarbons (HFC's). Industry in general, and the heat transfer and blowing agent segments of industry in particular are continually seeking new fluorocarbon based mixtures that offer alternatives to, and are considered environmentally safer substitutes for, CFCs and HCFCs. It is generally considered important, in many cases, however, that any potential substitute must also possess those properties present in many of the most widely used of such materials, such as excellent heat transfer properties, appropriate chemical stability, low- or no-toxicity, non-flammability and/or lubricant compatibility, among others, and other desirable foam characteristics when used as blowing agents.

[0007] Applicants have come to appreciate that lubricant compatibility is of particular importance in many of applications. More particularly, it is highly desirable for refrigeration fluids to be compatible with the lubricant utilized in the compressor unit, used in most refrigeration systems. Unfortunately, many non-chlorine-containing refrigeration fluids, including HFC's, are relatively insoluble and/or immiscible in the types of lubricants used traditionally with CFC's and HFC's, including, for example, mineral oils, alkylbenzenes or poly(alpha-olefins). In order for a refrigeration fluid/lubricant combination to work at a desirable level of efficiency within a compression refrigeration, air-conditioning and/or heat pump system, the lubricant should be sufficiently soluble in the refrigeration liquid over a wide range of operating temperatures. Such solubility lowers the viscosity of the lubricant and allows it to flow more easily throughout the system. In the absence of such solubility, lubricants tend to become lodged in the coils of the evaporator of the refrigeration, air-conditioning or heat pump system, as well as other parts of the system, and thus reduce the system efficiency.

[0008] With regard to efficiency in use, it is important to note that a loss in refrigerant thermodynamic performance or energy efficiency may have secondary environmental impacts

through increased fossil fuel usage arising from an increased demand for electrical energy.

[0009] Furthermore, it is generally considered desirably for CFC refrigerant and blowing agent substitutes to be effective without major engineering changes to conventional systems, such as vapor compression technology and foam generating systems.

[0010] Methods and compositions for making conventional foamed materials, such as for example thermoplastic materials and thermosetting materials, have long been known. These methods and compositions have typically utilized chemical and/or physical blowing agents to form the foamed structure in a polymeric matrix. Such blowing agents have included, for example, azo compounds, various volatile organic compounds (VOCs) and chlorofluorocarbons (CFCs). The chemical blowing agents typically undergo some form of chemical change, including chemical reaction with the material that forms the polymer matrix (usually at a predetermined temperature/pressure) that causes the release of a gas, such as nitrogen, carbon dioxide, or carbon monoxide. One of the most frequently used chemical blowing agents is water. The physical blowing agents typically are dissolved in the polymer or polymer precursor material and then expand volumetrically (again at a predetermined temperature/pressure) to contribute to the formation of the foamed structure. Physical blowing agents are frequently used in connection with thermoplastic foams, although chemical blowing agents can be used in place of or in addition to physical blowing agents in connection with thermoplastic foam. For example, it is known to use chemical blowing agents in connection with the formation of polyvinylchloride-based foams. It is common to use chemical blowing and/or physical blowing agents in connection with thermosetting foams. Of course, it is possible that certain compounds and the compositions that contain them may at once constitute a chemical and a physical blowing agent.

[0011] It was common in the past that the CFCs were used as standard blowing agents in the preparation of isocyanate-based foams, such as rigid and flexible polyurethane and polyisocyanurate foams. For example, compositions consisting of CFC materials, such as CC13F (CFC-11) had become a standard blowing agent. However, the use of this material has been banned by international treaty on the grounds that its release into the atmosphere damages the ozone layer in the stratosphere. As a consequence, it is no longer generally common that neat CFC-11 is used as a standard blowing agent for forming thermosetting foams, such as isocyanate-based foams and phenolic foams.

Flammability is another important property for many applications. That is, it is considered either important or essential in many applications, including particularly in heat transfer and blowing agent applications, to use compositions which are of low flammability or are non-flammable. Thus, it is frequently beneficial to use in such compositions compounds which are nonflammable. As used herein, the term “nonflammable” refers to compounds or compositions which are determined to be nonflammable as determined in accordance with ASTM standard E-681, dated 2002, which is incorporated herein by reference. Unfortunately, many HFC's which might otherwise be desirable for used in refrigerant or foam blowing agent compositions are not nonflammable. For example, fluoroalkane difluoroethane (HFC-152a) and the fluoroalkene 1,1,1-trifluoropropene (HFO-1243zf) are each flammable and therefore not viable for use in many applications.

Higher fluoroalkenes, that is fluorine-substituted alkenes having at least five carbon atoms, have been suggested for use as refrigerants. U.S. Pat. No. 4,788,352—Smutny is directed to production of fluorinated C5 to C8 compounds having at least some degree of unsaturation. The Smutny patent identifies such higher olefins as being known to have utility as refrigerants, pesticides, dielectric fluids, heat transfer fluids, solvents, and intermediates in various chemical reactions. (See column 1, lines 11-22).

[0012] Another example of a relatively flammable material is the fluorinated ether 1,1,2,2-tetrafluoroethyl methyl ether (which is referred to as HFE-254pc or also sometimes as HFE-254cb), which has been measured to have a flammability limit (vol %) of from about 5.4% to about 24.4%. Fluorinated ethers of this general type have been disclosed for use as blowing agents in U.S. Pat.

No. 5,137,932—Beheme et al, which is incorporated herein by reference.

[0013] It has been suggested to use bromine-containing halocarbon additives to decrease flammability of certain materials, including foam blowing agents, in U.S. Pat. No. 5,900,185—Tapscott. The additives in this patent are said to be characterized by high efficiency and short atmospheric lifetimes, that is, low ozone depletion potential (ODP) and a low global warming potential (GWP).

[0014] It is believed that the olefins described in Smutny and Tapscott have certain disadvantages. For example, some of these compounds may tend to attack substrates, particularly general-purpose plastics such as acrylic resins and ABS resins.

Furthermore, the higher olefinic compounds described in Smutny may also be undesirable in certain applications because of the potential level of toxicity of such compounds which may arise as a result of pesticide activity noted in Smutny. Also, such compounds may have a boiling point which is too high to make them useful as a refrigerant in certain applications.

[0015] Bromofluoromethane and bromochlorofluoromethane derivatives, particularly bromotrifluoromethane (Halon 1301) and bromochlorodifluoromethane (Halon 1211) have gained widespread use as fire extinguishing agents in enclosed areas such as airplane cabins and computer rooms. However, the use of various halons is being phased out due to their high ozone depletion. Moreover, as halons are frequently used in areas where humans are present, suitable replacements must also be safe to humans at concentrations necessary to suppress or extinguish fire.

[0016] Applicants have thus come to appreciate a need for compositions, and particularly heat transfer compositions, fire extinguishing/suppression compositions, blowing agents, solvent compositions, propellants, cleaning compositions and compatibilizing agents, that are potentially useful in numerous applications, including vapor compression heating and cooling systems and methods, while avoiding one or more of the disadvantages noted above.

SUMMARY OF THE INVENTION

[0017] Applicants have found that the above-noted need, and other needs, can be satisfied by compositions, including heat transfer compositions, blowing agent compositions, foams and foam premixes, solvent compositions, propellants, cleaning compositions and compatibilizing agents comprising one or more monochlorotrifluoropropene, preferably a compound selected from the group consisting of: [0018] $\text{transCF}_3\text{CH}=\text{CClH}$ (1233zdE); [0019] $\text{cisCF}_3\text{CH}=\text{CClH}$ (1233zdZ); [0020] $\text{transCHF}_2\text{CF}=\text{CClH}$ (1233ydE); [0021] $\text{cisCHF}_2\text{CF}=\text{CClH}$ (1233ydZ); [0022] $\text{transCHF}_2\text{CH}=\text{CClF}$ (1233zbE); [0023] $\text{cisCHF}_2\text{CH}=\text{CClF}$ (1233zbZ); [0024] $\text{transCHF}_2\text{CCl}=\text{CHF}$ (1233xeE); [0025] $\text{cisCHF}_2\text{CCl}=\text{CHF}$ (1233xeZ); [0026] $\text{CH}_2\text{FCCl}=\text{CF}_2$ (1233x0); [0027] $\text{transCHFC1CF}=\text{CFH}$ (1233yeE); [0028] $\text{cisCHFC1CF}=\text{CFH}$ (1233yeZ); [0029] $\text{CH}_2\text{C1CF}=\text{CF}_2$ (1233yc); [0030] $\text{CF}_2\text{C1CF}=\text{CH}_2$ (1233xf);

and combinations of two or more of these.

[0031] It is contemplated that all such compounds as identified above are adaptable for use in certain aspects of the present invention. The preferred compound(s) in accordance with the compositions and methods of the present invention preferably exhibit one or more, and preferably all, of the following properties: chemical stability; no substantial ozone depleting potential (ODP); relatively high degree of miscibility with common contaminants, particularly mineral oil and/or silicone oil; low or no flammability; low or no toxicity; and low or no global warming potential (GWP).

[0032] The preferred compounds for use in the present compositions have been found to possess at once several of these desirable beneficial properties. More specifically, the preferred compounds have: no substantial ozone depletion potential, preferably an ODP of not greater than about 0.5 and even more preferably of not greater than about 0.25, most preferably of not greater than about 0.1; a GWP of not greater than about 150, and even more preferably of not greater than about 50. In many of the preferred embodiments, the compound of the present invention has a normal boiling

point of from about 10° C. to about 60° C., and even more preferably of from about 15° C. to about 50° C., and even more preferably from about 10° C. to about 25° C. It is also generally preferred that the compound(s) have no flash point as measured by one of the standard flash point methods, for example ASTM-1310-86 "Flash point of liquids by tag Open-cup apparatus" and an atmospheric lifetime of not greater than about 100 days and even more preferably of not greater than about 50 days. Also, the preferred compound(s) are miscible with greater than 20% by weight of mineral oil and/or silicone oil, more preferably in a weight ratio in the range of at least about 80:20 to about 20:80, and even more preferably in substantially all proportions.

[0033] Preferred compounds of the present invention exhibit a relatively low toxicity value. As used herein, ODP is defined in the "Scientific Assessment of Ozone Depletion, 2002", a report of the World Meteorological association, incorporated here by reference. As used herein, GWP is defined relative to that of carbon dioxide and over a 100 year time horizon, and defined in the same reference as for the ODP mentioned above. As used herein, miscibility is measured in accordance with visual evaluation of phase formation or separation when two liquids are mixed together, as is known to those skilled in the art.

[0034] The compositions of the present invention thus generally possess properties and characteristics that are highly desirable for use in connection with many different applications, including many different types of cleaning and contaminant removal applications.

[0035] In certain embodiments, the monochlorotrifluoropropenes of the present invention may be used in combination with one or more other fluorinated olefins (hereinafter referred to for convenience but not by way of limitation as "additional fluoroalkene") having Formula I as follows:

XCFA3-z

where X is a C2, C3, C4 or C.sub.5 unsaturated, substituted or unsubstituted, radical, each R is independently Cl, F, Br, I or H, and z is 1 to 3, provided such compound is not a monochlorotrifluoropropene. In certain preferred embodiments, the additional fluoroalkene of the present invention has at least four (4) halogen substituents, at least three of which are F. Preferably in certain embodiments none of the substituents are Br. In certain preferred embodiments, the compound of Formula I comprises a compound, and preferably a three carbon compound, in which each non-terminal unsaturated carbon has at least one halogen substituent, more preferably at least one substituent selected from chlorine and fluorine, with compounds having at least three fluorines being especially preferred in certain embodiments.

[0036] In certain preferred embodiments, especially embodiments involving heat transfer compositions, blowing agent compositions, solvent compositions and cleaning compositions, the compound of Formula I is a three carbon olefin in which z is 1 or 2. Thus, the compound of Formula I in certain embodiments comprises a compound of Formula (IA):

$$CF''H_{2.w} = CR - CF_{zR3.z} \quad (IA) \quad [0037]$$
 where each R is independently Cl, F, Br, I or H, w is 1 or 2, and z is 1 or 2.

[0038] In certain preferred compounds of Formula IA each R is F or H, examples of which are:

[0039] $CF_{2.2} = CF - CH_{2.2}F$ (HF0-1234yc); [0040] $CF_2 = CH - CF_2H$ (HF0-1234zc); [0041] trans- $CHF = CF - CF_{2.2}H$ (HFO-1234ye(E)); and [0042] cis- $CHF = CF - CF_{2.2}H$ (HFO-1234ye(Z)).

[0043] For embodiments of Formula (IA) in which at least one Br substituent is present, it is preferred that the compound includes no hydrogen. In such embodiments it is also generally preferred that the Br substituent is on an unsaturated carbon, and even more preferably the Br substituent is on a non-terminal unsaturated carbon. One particularly preferred embodiment in this class is $CF_3CBr = CF_2$, including all of its isomers.

[0044] In certain embodiments it is highly preferred that the additional fluoroalkene compounds of

Formula I comprise propenes, butenes, pentenes and hexenes having from 3 to 5 fluorine substituents, with other substituents being either present or not present. In certain preferred embodiments, no R is Br, and preferably the unsaturated radical contains no Br substituents. Among the propenes, tetrafluoropropenes (HFO-1234) are especially preferred in certain embodiments.

[0045] In certain embodiments, pentafluoropropenes are preferred, including particularly those pentafluoropropenes in which there is a hydrogen substituent on the terminal unsaturated carbon, such as $\text{CF}_3\text{CF}=\text{CFH}$ (HFO-1225yeZ and/or yeE), particularly since applicants have discovered that such compounds have a relatively low degree of toxicity in comparison to at least the compound $\text{CF}_3\text{CH}=\text{CF}_2$ (HFO-1225w).

[0046] Among the butenes, fluorochlorobutenes are especially preferred in certain embodiments.

[0047] The term “HFO-1234” is used herein to refer to all tetrafluoropropenes. Among the tetrafluoropropenes are included 1,1,1,2-tetrafluoropropene (HFO-1234y0, both cis- and trans-1,1,1,3-tetrafluoropropene (HFO-1234ze), $\text{CF}_2=\text{CF}-\text{CH}_2\text{F}$ (HFO-1234yc), $\text{CF}_2=\text{CH}-\text{CF}_2\text{H}$ (HFO-1234zc), trans- $\text{CHF}=\text{CF}-\text{CF}_2\text{H}$ (HFO-1234ye(E)), and cis- $\text{CHF}=\text{CF}-\text{CF}_2\text{H}$ (HFO-1234ye(Z)). The term HFO-1234ze is used herein generically to refer to 1,1,1,3-tetrafluoropropene, independent of whether it is the cis- or trans-form. The terms “cisHFO-1234ze” and “transHFO-1234ze” are used herein to describe the cis- and trans-forms of 1,1,1,3-tetrafluoropropene respectively. The term “HFO-1234ze” therefore includes within its scope cisHFO-1234ze, transHFO-1234ze, and all combinations and mixtures of these. The term HFO-1234ye is used herein generically to refer to 1,2,3,3-tetrafluoropropene ($\text{CHF}=\text{CF}-\text{CF}_2\text{H}$), independent of whether it is the cis- or trans-form. The terms “cisHFO-1234ye” and “transHFO-1234ye” are used herein to describe the cis- and trans-forms of 1,2,3,3-tetrafluoropropene, respectively. The term “HFO-1234ye” therefore includes within its scope cisHFO-1234ye, transHFO-1234ye, and all combinations and mixtures of these.

[0048] The term “HFO-1225” is used herein to refer to all pentafluoropropenes. Among such molecules are included 1,1,1,2,3 pentafluoropropene (HFO-1225yez), both cis- and trans-forms thereof. The term HFO-1225yez is thus used herein generically to refer to 1,1,1,2,3 pentafluoropropene, independent of whether it is the cis- or trans-form. The term “HFO-1225yez” therefore includes within its scope cisHFO-1225yez, transHFO-1225yez, and all combinations and mixtures of these.

[0049] In certain preferred embodiments the composition comprises at least one monochlorotrifluoropropene compound and at least one additional fluorinated olefin, including tetrafluoropropene, with each being present in the composition in an amount of from about 20% by weight to about 80% by weight, more preferably from about 30% by weight to about 70% by weight, and even more preferably from about 40% by weight to about 60% by weight.

[0050] The present invention provides also methods and systems which utilize the compositions of the present invention. In one aspect the methods include methods and systems, for heat transfer, for retrofitting existing heat transfer equipment, and for replacing the existing heat transfer fluids in an existing heat transfer system. In other aspects the present compositions are used in connection with foams, foam blowing, forming foams and foam premixes, solvating, cleaning, flavor and fragrance extraction and/or delivery, aerosol generation, non-aerosol propellants and as inflating agents.

Description

DESCRIPTION OF THE DRAWINGS

[0051] FIG. 1 provides a sketch of the vessel as discussed in Example 4.

DETAILED DESCRIPTION OF THE INVENTION

A. The Compositions

[0052] The present compositions are believed to possess properties that are advantageous for a number of important reasons. For example, applicants believe, based at least in part on mathematical modeling, that the preferred compositions of the present invention will not have a substantial negative affect on atmospheric chemistry, being negligible contributors to ozone depletion in comparison to some other halogenated species. The preferred compositions of the present invention thus have the advantage of not contributing substantially to ozone depletion. The preferred compositions also do not contribute substantially to global warming compared to many of the hydrofluoroalkanes presently in use.

[0053] Of course other compounds and/or components that modulate a particular property of the compositions (such as cost for example) may also be included in the present compositions, and the presence of all such compounds and components is within the broad scope of the invention.

[0054] In certain preferred forms, compositions of the present invention have a Global Warming Potential (GWP) of not greater than about 1500, more preferably not greater than about 1000, more preferably not greater than about 500, and even more preferably not greater than about 150. In certain embodiments, the GWP of the present compositions is not greater than about 100 and even more preferably not greater than about 75. As used herein, "GWP" is measured relative to that of carbon dioxide and over a 100 year time horizon, as defined in "The Scientific Assessment of Ozone Depletion, 2002, a report of the World Meteorological Association's Global Ozone Research and Monitoring Project," which is incorporated herein by reference.

[0055] In certain preferred forms, the present compositions also preferably have an Ozone Depletion Potential (ODP) of not greater than 0.05, more preferably not greater than 0.02 and even more preferably about zero. As used herein, "ODP" is as defined in "The Scientific Assessment of Ozone Depletion, 2002, A report of the World Meteorological Association's Global Ozone Research and Monitoring Project," which is incorporated herein by reference.

[0056] The amount of the monochlorotrifluoropropene contained in the present compositions can vary widely, depending the particular application, and compositions containing more than trace amounts and less than 100% of the compound are within broad the scope of the present invention. Moreover, the compositions of the present invention can be azeotropic, azeotrope-like or non-azeotropic.

[0057] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise transCF₃CH=CClH (1233zdE) in amounts from about 5% by weight to about 99% by weight, and even more preferably from about 5% to about 95%.

[0058] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise cisCF₃CH=CClH (1233zdZ) in amounts from about 5% by weight to about 99% by weight, and even more preferably from about 5% to about 95%.

[0059] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise transCHF₂CF=CClH (1233ydE) in amounts from about 5% by weight to about 99% by weight, and even more preferably from about 5% to about 95%.

[0060] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise cisCHF₂CF=CClH (1233ydZ) in amounts from about 5% by weight to about 99% by weight, and even more preferably from about 5% to about 95%.

[0061] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise transCHFClCF=CFH (1233yeE) in amounts from about 5% by weight to about 99% by weight, and even more preferably from about 5% to about 95%.

[0062] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise cisCHFClCF=CFH (1233yeZ) in amounts from about 5% by weight to about 99% by weight, and even more preferably from about 5% to about 95%.

[0063] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise transCF₃CH=CClH (1233zbE) in an amount that is at least about 50% by weight, and even more preferably at least about 70% by weight, of the composition.

[0064] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise cisCF.sub.3CH—CC1H (1233ybZ) in an amount that is at least about 50% by weight, and even more preferably at least about 70% by weight, of the composition.

[0065] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise transCHF.sub.2CF=CC1H (1233ydE) in an amount that is at least about 50% by weight, and even more preferably at least about 70% by weight, of the composition.

[0066] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise cisCHF.sub.2CF=CC1H (1233ydZ) in an amount that is at least about 50% by weight, and even more preferably at least about 70% by weight, of the composition.

[0067] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise transCHFC1CF=CFH (1233yeE) in an amount that is at least about 50% by weight, and even more preferably at least about 70% by weight, of the composition.

[0068] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise cisCHFC1CF=CFH (1233yeZ) in an amount that is at least about 50% by weight, and even more preferably at least about 70% by weight, of the composition.

[0069] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise CH2C1CF=CF2 (1233cf) in an amount that is at least about 50% by weight, and even more preferably at least about 70% by weight, of the composition.

[0070] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise CF2C1CF=CH.sub.2 (1233y0) in an amount that is at least about 50% by weight, and even more preferably at least about 70% by weight, of the composition.

[0071] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise transCHF2CC1=CHF (1233xeE) in an amount that is at least about 50% by weight, and even more preferably at least about 70% by weight, of the composition.

[0072] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise cis CHF2CC1=CHF (1233xeZ) in an amount that is at least about 50% by weight, and even more preferably at least about 70% by weight, of the composition.

[0073] In certain preferred embodiments, the present compositions, particularly blowing agent and heat transfer compositions comprise CH2FCC1=CF2 (1233xc) in an amount that is at least about 50% by weight, and even more preferably at least about 70% by weight, of the composition.

[0074] Many additional compounds or components, including lubricants, stabilizers, metal passivators, corrosion inhibitors, flammability suppressants, and other compounds and/or components that modulate a particular property of the compositions (such as cost for example) may be included in the present compositions, and the presence of all such compounds and components is within the broad scope of the invention. In certain preferred embodiments, the present compositions include, in addition to the one or more monochlorotrifluoropropene compounds mentioned above, one or more of the following: [0075] trichlorofluoromethane (CFC-11); [0076] Dichlorodifluoromethane (CFC-12); [0077] difluoromethane (HFC-32); [0078] pentafluoroethane (HFC-125); [0079] 1,1,2,2-tetrafluoroethane (HFC-134); [0080] 1,1,1,2-Tetrafluoroethane (HFC-134a); [0081] Difluoroethane (HFC-152a); [0082] 1,1,1,2,3,3,3-heptafluoropropane (HFC-227 ea); [0083] 1,1,1,3,3,3-hexafluoropropane (HFC-236fa); [0084] 1,1,1,3,3-pentafluoropropane (HFC-245fa); [0085] 1,1,1,3,3-pentafluorobutane (HFC-365mfc); [0086] water; and [0087] CO.sub.2.

[0088] The relative amount of any of the above noted compounds of the present invention, as well as any additional components which may be included in present compositions, can vary widely within the general broad scope of the present invention according to the particular application for the composition, and all such relative amounts are considered to be within the scope thereof.

[0089] Accordingly, applicants have recognized that certain compositions of the present invention

can be used to great advantage in a number of applications. For example, included in the present invention are methods and compositions relating to heat transfer applications, foam and blowing agent applications, propellant applications, sprayable composition applications, sterilization applications, aerosol applications, compatibilizer application, fragrance and flavor applications, solvent applications, cleaning applications, inflating agent applications and others. It is believed that those of skill in the art will be readily able to adapt the present compositions for use in any and all such applications without undue experimentation.

[0090] The present compositions are generally useful as replacements for CFCs, such as dichlorodifluoromethane (CFC-12), HCFCs, such as chlorodifluoromethane (HCFC-22), HFCs, such as tetrafluoroethane (HFC-134a), and combinations of HFCs and CFCs, such as the combination of CFC-12 and 1,1-difluoroethane (HFC-152a) (the combination CFC-12:HFC-152a in a 73.8:26.2 mass ratio being known as R-500) in refrigerant, aerosol, and other applications.

B. Heat Transfer Applications

[0091] The compositions of the present invention are generally adaptable for use in heat transfer applications, that is, as a heating and/or cooling medium, including as evaporative cooling agents.

[0092] In connection with evaporative cooling applications, the compositions of the present invention are brought in contact, either directly or indirectly, with a body to be cooled and thereafter permitted to evaporate or boil while in such contact, with the preferred result that the boiling gas in accordance with the present composition absorbs heat from the body to be cooled. In such applications it may be preferred to utilize the compositions of the present invention, preferably in liquid form, by spraying or otherwise applying the liquid to the body to be cooled. In other evaporative cooling applications, it may be preferred to permit a liquid composition in accordance with the present invention to escape from a relatively high pressure container into a relatively lower pressure environment wherein the body to be cooled is in contact, either directly or indirectly, with the container enclosing the liquid composition of the present invention, preferably without recovering or recompressing the escaped gas. One particular application for this type of embodiment is the self-cooling of a beverage, food item, novelty item or the like. Previous to the invention described herein, prior compositions, such as HFC-152a and HFC-134a were used for such applications. However, such compositions have recently been looked upon negatively in such application because of the negative environmental impact caused by release of these materials into the atmosphere. For example, the United States EPA has determined that the use of such prior chemicals in this application is unacceptable due to the high global warming nature of these chemicals and the resulting detrimental effect on the environment that may result from their use. The compositions of the present invention should have a distinct advantage in this regard due to their low global warming potential and low ozone depletion potential, as described herein.

Additionally, the present compositions are expected to also find substantial utility in connection with the cooling of electrical or electronic components, either during manufacture or during accelerated lifetime testing. In a accelerated lifetime testing, the component is sequentially heated and cooled in rapid succession to simulate the use of the component. Such uses would therefore be of particular advantage in the semiconductor and computer board manufacturing industry. Another advantage of the present compositions in this regard is they are expected to exhibit as contagious electrical properties when used in connection with such applications. Another evaporative cooling application comprises methods for temporarily causing a discontinuation of the flow of fluid through a conduit. Preferably, such methods would include contacting the conduit, such as a water pipe through which water is flowing, with a liquid composition according to the present invention and allowing the liquid composition of the present invention to evaporate while in contact with the conduit so as to freeze liquid contained therein and thereby temporarily stop the flow of fluid through the conduit. Such methods have distinct advantage in connection with enabling the service or other work to be performed on such conduits, or systems connected to such conduits, at a location downstream of the location at which the present composition is applied.

[0093] The relative amount of the hydrofluoroolefin used in accordance with the present invention is preferably selected to produce a heat transfer fluid which has the required heat transfer capacity, particularly refrigeration capacity, and preferably is at the same time nonflammable. As used herein, the term non-flammable refers to a fluid which is non-flammable in all proportions in air as measured by ASTM E-681.

[0094] The compositions of the present invention may include other components for the purpose of enhancing or providing certain functionality to the composition, or in some cases to reduce the cost of the composition. For example, refrigerant compositions according to the present invention, especially those used in vapor compression systems, include a lubricant, generally in amounts of from about 30 to about 50 percent by weight of the composition. Furthermore, the present compositions may also include a co-refrigerant, or compatibilizer, such as propane, for the purpose of aiding compatibility and/or solubility of the lubricant. Such compatibilizers, including propane, butanes and pentanes, are preferably present in amounts of from about 0.5 to about 5 percent by weight of the composition. Combinations of surfactants and solubilizing agents may also be added to the present compositions to aid oil solubility, as disclosed by U.S. Pat. No. 6,516,837, the disclosure of which is incorporated by reference. Commonly used refrigeration lubricants such as Polyol Esters (POEs) and Poly Alkylene Glycols (PAGs), PAG oils, silicone oil, mineral oil, alkyl benzenes (ABs) and poly(alpha-olefin) (PAO) that are used in refrigeration machinery with hydrofluorocarbon (HFC) refrigerants may be used with the refrigerant compositions of the present invention. Commercially available mineral oils include Witco LP 250 (registered trademark) from Witco, Zerol 300 (registered trademark) from Shrieve Chemical, Sunisco 3GS from Witco, and Calumet R015 from Calumet. Commercially available alkyl benzene lubricants include Zerol 150 (registered trademark). Commercially available esters include neopentyl glycol dipelargonate, which is available as Emery 2917 (registered trademark) and Hatcol 2370 (registered trademark). Other useful esters include phosphate esters, dibasic acid esters, and fluoroesters. In some cases, hydrocarbon based oils are have sufficient solubility with the refrigerant that is comprised of an iodocarbon, the combination of the iodocarbon and the hydrocarbon oil might more stable than other types of lubricant. Such combination may therefore be advantageous. Preferred lubricants include polyalkylene glycols and esters. Polyalkylene glycols are highly preferred in certain embodiments because they are currently in use in particular applications such as mobile air-conditioning. Of course, different mixtures of different types of lubricants may be used.

[0095] In certain preferred embodiments, the heat transfer composition comprises from about 10% to about 95% by weight of one or more monochlorotrifluorpropene as described above, and from about 5% to about 90% by weight of an adjuvant, particular in certain embodiments a co-refrigerant (such as HFC-152, HFC-125 and/or CF.sub.3I). The use of the term co-refrigerant is not intended for use herein in a limiting sense regarding the relative performance of the monochlorotrifluorpropene compound, but is used instead used to identify other components of the refrigerant composition generally that contribute to the desirable heat transfer characteristics of the composition for a desired application. In certain of such embodiments the co-refrigerant comprises, and preferably consists essentially of, one or more HFCs and/or one or more fluoroiodo C1-C3 compounds, such as trifluoroiodomethane, and combinations of these with each other and with other components.

[0096] In preferred embodiments in which the co-refrigerant comprises HFC, preferably HFC-125 the composition comprises HFC in an amount of from about 50% by weight to about 95% by weight of the total heat transfer composition, more preferably from about 60% by weight to about 90% by weight, and even more preferably of from about 70% to about 90% by weight of the composition. In such embodiments the monochlorotrifluorpropene(s) compound of the present invention preferably comprises from about 5% by weight to about 50% by weight of the total heat transfer composition, more preferably from about 10% by weight to about 40% by weight, and even more preferably of from about 10% to about 30% by weight of the composition.

[0097] In preferred embodiments in which the co-refrigerant comprises fluoroiodocarbon, preferably CF₃I, the composition comprises fluoroiodocarbon in an amount of from about 15% by weight to about 50% by weight of the total heat transfer composition, more preferably from about 20% by weight to about 40% by weight, and even more preferably of from about 25% to about 35% by weight of the composition. In such embodiments the monochlorotrifluoropropene(s) compound of the present invention preferably comprises an amount of from about 50% by weight to about 90% by weight of the total heat transfer composition, more preferably from about 60% by weight to about 80% by weight, and even more preferably of from about 65% to about 75% by weight of the composition.

[0098] The present methods, systems and compositions are thus adaptable for use in connection with a wide variety of heat transfer systems in general and refrigeration systems in particular, such as air-conditioning (including both stationary and mobile air conditioning systems), refrigeration, heat-pump systems, and the like. In certain preferred embodiments, the compositions of the present invention are used in refrigeration systems originally designed for use with an HFC refrigerant, such as, for example, HFC-134a, or an HCFC refrigerant, such as, for example, HCFC-22. The preferred compositions of the present invention tend to exhibit many of the desirable characteristics of HFC-134a and other HFC refrigerants, including a GWP that is as low, or lower than that of conventional HFC refrigerants and a capacity that is as high or higher than such refrigerants and a capacity that is substantially similar to or substantially matches, and preferably is as high as or higher than such refrigerants. In particular, applicants have recognized that certain preferred embodiments of the present compositions tend to exhibit relatively low global warming potentials ("GWPs"), preferably less than about 1000, more preferably less than about 500, and even more preferably less than about 150. In addition, the relatively constant boiling nature of certain of the present compositions, including the azeotrope-like compositions described in the co-pending patent applications incorporated herein by reference, makes them even more desirable than certain conventional HFCs, such as R-404A or combinations of HFC-32, HFC-125 and HFC-134a (the combination HFC-32:HFC-125:HFC134a in approximate 23:25:52 weight ratio is referred to as R-407C), for use as refrigerants in many applications. Heat transfer compositions of the present invention are particularly preferred as replacements for HFC-134, HFC-152a, HFC-22, R-12 and R-500.

[0099] In certain other preferred embodiments, the present compositions are used in refrigeration systems originally designed for use with a CFC-refrigerant. Preferred refrigeration compositions of the present invention may be used in refrigeration systems containing a lubricant used conventionally with CFC-refrigerants, such as mineral oils, polyalkylbenzene, polyalkylene glycol oils, and the like, or may be used with other lubricants traditionally used with HFC refrigerants. As used herein the term "refrigeration system" refers generally to any system or apparatus, or any part or portion of such a system or apparatus, which employs a refrigerant to provide cooling. Such refrigeration systems include, for example, air conditioners, electric refrigerators, chillers (including chillers using centrifugal compressors), transport refrigeration systems, commercial refrigeration systems and the like.

[0100] Many existing refrigeration systems are currently adapted for use in connection with existing refrigerants, and the compositions of the present invention are believed to be adaptable for use in many of such systems, either with or without system modification. Many applications the compositions of the present invention may provide an advantage as a replacement in smaller systems currently based on certain refrigerants, for example those requiring a small refrigerating capacity and thereby dictating a need for relatively small compressor displacements. Furthermore, in embodiments where it is desired to use a lower capacity refrigerant composition of the present invention, for reasons of efficiency for example, to replace a refrigerant of higher capacity, such embodiments of the present compositions provide a potential advantage. Thus, it is preferred in certain embodiments to use compositions of the present invention, particularly compositions

comprising a substantial proportion of, and in some embodiments consisting essentially of the present compositions, as a replacement for existing refrigerants, such as: HFC-134a; CFC-12; HCFC-22; HFC-152a; combinations of pentafluoroethane (HFC-125), trifluoroethane (HFC-143a) and tetrafluoroethane (HFC-134a) (the combination HFC-125:HFC-143a:HFC-134a in approximate 44:52:4 weight ratio is referred to as R-404A); combinations of HFC-32, HFC-125 and HFC-134a (the combination HFC-32:HFC-125:HFC-134a in approximate 23:25:52 weight ratio is referred to as R-407C); combinations of methylene fluoride (HFC-32) and pentafluoroethane (HFC-125) (the combination HFC-32:HFC-125 in approximate 50:50 weight ratio is referred to as R-410A); the combination of CFC-12 and 1,1-difluoroethane (HFC-152a) (the combination CFC-12:HFC-152a in a 73.8:26.2 weight ratio is referred to as R-500); and combinations of HFC-125 and HFC-143a (the combination HFC-125:HFC-143a in approximate 50:50 weight ratio is referred to as R-507A). In certain embodiments it may also be beneficial to use the present compositions in connection with the replacement of refrigerants formed from the combination HFC-32:HFC-125:HFC-134a in approximate 20:40:40 weight ratio, which is referred to as R-407A, or in approximate 15:15:70 weight ratio, which is referred to as R-407D. The present compositions are also believed to be suitable as replacements for the above noted compositions in other applications, such as aerosols, blowing agents and the like, as explained elsewhere herein.

[0101] In certain applications, the refrigerants of the present invention potentially permit the beneficial use of larger displacement compressors, thereby resulting in better energy efficiency than other refrigerants, such as HFC-134a. Therefore, the refrigerant compositions of the present invention provide the possibility of achieving a competitive advantage on an energy basis for refrigerant replacement applications, including automotive air conditioning systems and devices, commercial refrigeration systems and devices, chillers, residential refrigerator and freezers, general air conditioning systems, heat pumps and the like.

[0102] Many existing refrigeration systems are currently adapted for use in connection with existing refrigerants, and the compositions of the present invention are believed to be adaptable for use in many of such systems, either with or without system modification. In many applications the compositions of the present invention may provide an advantage as a replacement in systems which are currently based on refrigerants having a relatively high capacity. Furthermore, in embodiments where it is desired to use a lower capacity refrigerant composition of the present invention, for reasons of cost for example, to replace a refrigerant of higher capacity, such embodiments of the present compositions provide a potential advantage. Thus, it is preferred in certain embodiments to use compositions of the present invention, particularly compositions comprising a substantial proportion of, and in some embodiments consisting essentially of, HFO-1233 as a replacement for existing refrigerants, such as HFC-134a. In certain applications, the refrigerants of the present invention potentially permit the beneficial use of larger displacement compressors, thereby resulting in better energy efficiency than other refrigerants, such as HFC-134a. Therefore, the refrigerant compositions of the present invention provide the possibility of achieving a competitive advantage on an energy basis for refrigerant replacement applications.

[0103] It is contemplated that the compositions of the present also have advantage (either in original systems or when used as a replacement for refrigerants such as CFC-11, CFC-12, HCFC-22, HFC-134a, HFC-152a, R-500 and R-507A), in chillers typically used in connection with commercial air conditioning systems. In certain of such embodiments it is preferred to include in the present compositions from about 0.5 to about 30% of a supplemental flammability suppressant, and in certain cases more preferably 0.5% to about 15% by weight and even more preferably from about 0.5 to about 10% on a weight basis.

C. Blowing Agents, Foams and Foamable Applications

[0104] Blowing agents may also comprise or constitute one or more of the present compositions. As mentioned above, the compositions of the present invention may include the compounds of the present invention in widely ranging amounts. It is generally preferred, however, that for preferred

compositions for use as blowing agents in accordance with the present invention, one or more of the monochlorotrifluoropropene compound(s) are present in an amount that is at least about 5% by weight, and even more preferably at least about 15% by weight, of the composition. In certain preferred embodiments, the blowing agent comprises at least about 50% by weight of the present compositions, and in certain embodiments the blowing agent consists essentially of the present compositions. In certain preferred embodiments, the blowing agent compositions of the present invention and include, in addition to the monochlorotrifluoropropene compound(s) one or more of co-blowing agents, fillers, vapor pressure modifiers, flame suppressants, stabilizers and like adjuvants. The co-blowing agent in accordance with the present invention can comprise a physical blowing agent, a chemical blowing agent (which preferably in certain embodiments comprises water) or a blowing agent having a combination of physical and chemical blowing agent properties. It will also be appreciated that the blowing agents included in the present compositions, including the compounds of Formula I as well as the co-blowing agent, may exhibit properties in addition to those required to be characterized as a blowing agent. For example, it is contemplated that the blowing agent compositions of the present invention may include components, including the compounds or Formula I described above, which also impart some beneficial property to the blowing agent composition or to the foamable composition to which it is added. For example, it is within the scope of the present invention for the compound of Formula I or for the co-blowing agent to also act as a polymer modifier or as a viscosity reduction modifier.

[0105] By way of example, one or more of the following components may be included in certain preferred blowing agents of the present invention in widely varying amounts: hydrocarbons, hydrofluorocarbons (HFCs), ethers, alcohols, aldehydes, ketones, methyl formate, formic acid, water, trans-1,2-dichloroethylene, carbon dioxide and combinations of any two or more of these. Among ethers, it is preferred in certain embodiments to use ethers having from one to six carbon atoms. Among alcohols, it is preferred in certain embodiments to use alcohols having from one to four carbon atoms. Among aldehydes, it is preferred in certain embodiments to use aldehydes having from one to four carbon atoms.

[0106] Certain of the co-agents available for use in accordance with the present invention are describe below.

1. The Ethers

[0107] In certain preferred embodiments, present compositions, particularly blowing agent compositions, include at least one ether, preferably which functions as a co-blowing agent in the composition.

[0108] The ether(s) used in accordance with this aspect of the invention comprise fluorinated ethers (FEs), more preferably one or more hydro-fluorinated ethers (HFEs)), and even more preferably one or more C3 to C5 hydro-fluorinated ethers in accordance with Formula (III) below:

C.sub.allbF.sub.c-0-Cali.sub.eFf (III) [0109] where [0110] a=1-6, more preferably 2-5, and even more preferably 3-5, [0111] b=1-12, more preferably 1-6, and even more preferably 3-6, [0112] c=1-12, more preferably 1-6, and even more preferably 2-6, [0113] d=1-2 [0114] e=0-5, more preferably 1-3 [0115] f=0-5, more preferably 0-2, [0116] and where one of said Ca may be bound to one of said Ca to form a cyclofluoroether.

[0117] Certain preferred embodiments of the present invention are directed to compositions comprising at least one fluoroalkene as described herein, preferably in certain embodiments chlorofluoroallcenes such as HFCO-1233xd, and at least one fluoro-ether, more preferably at least one hydro-fluoroether, containing from 2 to 8, preferably 2 to 7, and even more preferably 2 to 6 carbon atoms, and in certain embodiments most preferably three carbon atoms. The hydro-fluoroether compounds of the present invention are sometimes referred to herein for the purpose of convenience as hydrofluoro-ethers or "HFEs" if they contain at least one hydrogen.

[0118] Applicants believe that, in general, the fluoroethers in accordance with the present

disclosure and in particular in accordance with above identified Formula (III) are generally effective and exhibit utility in combination with the fluoroalkene compounds in accordance with the teachings contained herein. However, applicants have found that from among the fluoroethers, it is preferred to use in certain embodiments, especially embodiments relating to blowing agent compositions and foam and foaming methods, to utilize hydrofluoroethers that are at least difluorinated, more preferably at least trifluorinated, and even more preferably at least tetrafluorinated. Especially preferred in certain embodiments are tetrafluorinated fluoroethers having from 3 to 5 carbon atoms, more preferably 3 to 4 carbon atoms, and even more preferably 3 carbon atoms.

[0119] In certain preferred embodiments, the ether compound of the present invention comprises a 1,1,2,2-tetrafluoroethylmethylether (which is sometimes referred to herein as HFE-245pc or HFE-245cb2), including any and all isomeric forms thereof.

[0120] The amount of the Formula III compounds, particularly 1,1,2,2-tetrafluoroethylmethylether contained in the present compositions can vary widely, depending the particular application, and compositions containing more than trace amounts and less than 100% of the compound are within broad the scope of the present invention. In preferred embodiments, the present compositions, particularly blowing agent compositions, comprise Formula III compounds, including preferred groups of compounds, in amounts from about 1% by weight to about 99% by weight, more preferably from about 5% to about 95% by weight, and even more preferably from 40% to about 90% by weight.

[0121] One or more of following compounds are preferred for use in accordance with certain preferred embodiments of the present invention: [0122] CHF₂OCH₂F (HFE-143E); [0123] CH₂FOCH₂F (HFE-152E); [0124] CH₂FOCH₂CF₃ (HFE-161E); [0125] cyclo-CF₂CH₂OCF₂O (HFE-c234fEa(3)); [0126] cyclo-CF₂CH₂OCF₂CH₂O (HFE-c234fE(3y)); [0127] CHF₂OCF₂CHF₂ (HFE-236caE); [0128] CF₃CF₂OCH₂F (HFE-236cbE(3'y)); [0129] CF₃OCF₂CHF₂ (HFE-236eaEa(3)); [0130] CHF₂OCF₂CH₂F (HFE-245eaEa(3)); [0131] CH₂FOCF₂CHF₂ (HFE-245caE(37)); [0132] CF₂OCF₂CH₂F (HFE-245cbE(3y)); [0133] CH₂FOCF₂CH₂F (HFE-245eaE(37)); [0134] CHF₂OCF₂CH₂F (HFE-245ebEa(3)); [0135] CF₃CHFOCH₂F (HFE-245ebE(37)); [0136] CF₃CHFOCH₂F (HFE-245ebE(37)); [0137] CF₃CHFOCH₂F (HFE-245faEaf(3)); [0138] CHF₂OCH₂CF₃ (HFE-245faE(37)); [0139] CH₂FCF₂OCH₂F (HFE-254caE); [0140] CHF₂OCF₂CH₃ (HFE-254cbEa(1)); [0141] CHF₂CF₂OCH₃ (HFE-254caE(3y)); [0142] CH₂FOCH₂CF₂CH₂F (HFE-254eaEct13); [0143] CF₃OCF₂CH₃ (HFE-254ebEa(3)); [0144] CF₂OCF₂CH₂CH₃ (HFE-254ebE(3y)); [0145] CHF₂OCF₂CH₂CH₃ (HFE-254faE); [0146] CF₂OCF₂CH₂CH₃ (HFE-254fbEct(3)); [0147] CF₃CH₂OCH₂F (HFE-254fbE137); [0148] CH₃OCF₂CH₂F (HFE-263caE(37)); [0149] CF₃CH₂OCH₃ (HFE-263fbE(3y)); [0150] CH₃OCH₂CHF₂ (HFE-272fbEf(3y)); [0151] CHF₂OCH₂CF₂CF₃ (HFE-338mceEy6); [0152] CHF₂OCF₂CH₂CF₃ (HFE-338mceEy6); [0153] CF₃CF₂OCH₂CF₃ (HFE-338mfE(3y)); [0154] (CF₃)₂CHOCH₂F (HFE-338mmzE(3y)); [0155] CF₃CF₂CF₂OCH₃ (HFE-347sEy8); [0156] CHF₂OCH₂CF₂CF₃ (HFE-347mfcEyS); [0157] CF₃OCH₂CF₂CHF₂ (HFE-347mfcEa(3)); [0158] CH₃OCF₂CH₂CF₃ (HFE-356mecE78); [0159] CH₃OCH(CF₃)₂ (HFE-356mmzE(3y)); [0160] CF₂OCF₂CH₂CH₃ (HFE-365mcEl37); [0161] CF₂OCF₂CH₂CH₃ (HFE-365mcEy6); [0162] CF₃CF₂CF₂OCH₂CF₃ (HFE-42-11meEy8); [0163] CF₂OCF₂CH₂CF₂CH₃; [0164] CF₃CF₂CF₂CF₂OCH₂CH₃; [0165] CF₃CF₂CF₂CF₂OCH₂CH₃; [0166] CF₃CF₂CF₂CF₂OCH₂CH₃; [0167] CF₃CF₂CF₂CF₂OCH₂CH₃; [0168] CF₃CF₂CF₂OCH₃.

[0149] It should be understood that the present inventors contemplate that any two or more of the above noted HFEs, may be used in combination in accordance with preferred aspects of the present invention. For example, it is contemplated that a material sold under the trade name HFE-7100 by 3M, which is understood to be a mixture of from about 20% to about 80% of methyl nonafluoroisobutyl ether and from about 20% to about 80% methyl nonafluorobutyl ether, may be used to advantage in accordance with certain preferred embodiments of the present invention. By

way of further example, it is contemplated that a material sold under the trade name HFE-7200 by 3M, which is understood to be a mixture of from about 20% to about 80% of ethyl nonafluoroisobutyl ether and from about 20% to about 80% ethyl nonafluorobutyl ether, may be used to advantage in accordance with certain preferred embodiments of the present invention. [0150] It is also contemplated that any one or more of the above-listed HFEs may be used in combination with other compounds as well, including other HFEs not specifically listed herein and/or other compounds with which the designated fluoroether is known to form an azeotrope. For example, each of the following compounds is known to form an azeotrope with trans-dichloroethylene, and it is contemplated that for the purposes of the present invention the use of such azeotropes should be considered to be within the broad scope of the invention: [0151] $\text{CF}_3\text{CFCF}_3\text{CF}_2\text{OCH}_3$; [0152] $\text{CF}_3\text{CF}_2\text{CF}_2\text{CF}_2\text{OCH}_3$; [0153] $\text{CF}_3\text{CFCF}_3\text{CF}_2\text{OCH}_2\text{CH}_3$; [0154] $\text{CF}_3\text{CF}_2\text{CF}_2\text{CF}_2\text{OCH}_2\text{CH}_3$; and [0155] $\text{CF}_3\text{CF}_2\text{CF}_2\text{OCH}_3$.

2. The Hydrofluorocarbons

[0156] In certain embodiments it is preferred that the compositions of the present invention, including particularly the blowing agent compositions of the present invention, include one or more HFCs as co-blowing agents, more preferably one or more C1-C4 HFCs. For example, the present blowing agent compositions may include one or more of difluoromethane (HFC-32), fluoroethane (HFC-161), difluoroethane (HFC-152), trifluoroethane (HFC-143), tetrafluoroethane (HFC-134), pentafluoroethane (HFC-125), pentafluoropropane (HFC-245), hexafluoropropane (HFC-236), heptafluoropropane (HFC-227ea), pentafluorobutane (HFC-365), hexafluorobutane (HFC-356) and all isomers of all such HFC's.

[0157] In certain embodiments, one or more of the following HFC isomers are preferred for use as co-blowing agents in the compositions of the present invention: [0158] fluoroethane (HFC-161); [0159] 1,1,1,2,2-pentafluoroethane (HFC-125); [0160] 1,1,2,2-tetrafluoroethane (HFC-134); [0161] 1,1,1,2-tetrafluoroethane (HFC-134a); [0162] 1,1,1-trifluoroethane (HFC-143a); [0163] 1,1-difluoroethane (HFC-152a); [0164] 1,1,1,2,3,3,3-heptafluoropropane (HFC-227ea); [0165] 1,1,1,3,3,3-hexafluoropropane (HFC-236fa); [0166] 1,1,1,2,3,3-hexafluoropropane (HFC-236ea); [0167] 1,1,1,2,3-pentafluoropropane (HFC-245eb); [0168] 1,1,2,2,3-pentafluoropropane (HFC-245ca); [0169] 1,1,1,3,3-pentafluoropropane (HFC-245fa); [0170] 1,1,1,3,3-pentafluorobutane (HFC-365mfc); and [0171] 1,1,1,2,2,3,4,5,5,5-decafluoropentane (HFC-43-10-mee).

3. The Hydrocarbons

[0172] In certain embodiments it is preferred that the compositions of the present invention, including particularly the blowing agent compositions of the present invention include one or more hydrocarbons, more preferably C3-C6 hydrocarbons. The present blowing agent compositions may include in certain preferred embodiments, for example: propane; iso- and normal-butane (each of such butanes being preferred for use as a blowing agent for thermoplastic foams); iso-, normal-, neo- and/or cyclo-pentane (each of such pentanes being preferable for use as a blowing agent for thermoset foams); iso- and normal-hexane; and heptanes.

[0173] Certain preferred embodiments of the present compositions, including particularly the blowing agent compositions, comprise one or more monochlorotrifluoropropenes, particularly HFC0-1233zd, and at least one hydrocarbon selected from the group consisting of iso-pentane, normal-pentane, cyclo-pentane, and combinations of these, with combinations comprising from about 50% to about 85% by weight of cyclo-pentane, and even more preferably from about 65% to about 75% by weight of cyclo-pentane, being preferred.

4. The Alcohols

[0174] In certain embodiments it is preferred that the compositions of the present invention, including particularly the blowing agent compositions of the present invention, include one or more alcohols, preferably one or more C1-C4 alcohols. For example, the present blowing agent compositions, the aerosol, the cleaning and the solvent compositions of the present invention may include one or more of methanol, ethanol, propanol, isopropanol, butanol, isobutanol, t-butanol,

and octanols. From among the octanols, isooctanol (i.e., 2-ethyl-1-hexanol) is preferred for use in blowing agent formulations and in solvent compositions

[0175] Certain preferred embodiments of the present compositions, including particularly the blowing agent compositions, comprise one or more monochlorotrifluoropropenes, particularly HFC0-1233zd, and at least one alcohol selected from the group consisting of methanol, ethanol, propanol, isopropanol, butanol, isobutanol, t-butanol and combinations of these.

5. The Aldehydes

[0176] In certain embodiments it is preferred that the compositions of the present invention, including particularly the blowing agent, the aerosol, the cleaning and the solvent compositions of the present invention include one or more aldehydes, particularly C1-C4 aldehydes, including formaldehyde, acetaldehyde, propanal, butanal and isobutanal.

6. The Ketones

[0177] In certain embodiments it is preferred that the compositions of the present invention, including particularly the blowing agent compositions, the aerosol, cleaning and the solvent compositions of the present invention include one or more ketones, preferably C1-C4 ketones. For example, the present blowing agent, the aerosol, the cleaning and the solvent compositions may include one or more of acetone, methylethylketone, and methylisobutylketone.

7. The Chlorocarbons

[0178] In certain embodiments it is preferred that compositions of the present invention, including particularly the blowing agent, the aerosol, the cleaning and the solvent compositions of the present invention include one or more chlorocarbons, more preferably C1-C3 chlorocarbons. The present compositions may include in certain preferred embodiments, for example: 1-chloropropane; 2-chloropropane; trichloroethylene; perchloroethylene; methene chloride; trans-1,2 dichloroethylenes, and combinations of these, with trans-1,2 dichloroethylenes being especially preferred in certain embodiments, particularly blowing agent embodiments.

8. Other Compounds

[0179] In certain embodiments it is preferred that compositions of the present invention, including particularly the blowing agent, the aerosol, the cleaning and the solvent compositions of the present invention include one or more additional compounds, including water, CO₂, methylformate, formic acid, dimethoxymethane (DME) and combinations of these. From among the above, DME is particularly preferred for use in blowing agent compositions and as a propellant in aerosol compositions in accordance with the present invention, particularly in combination with HFC0-1233zd. From among the above, water and CO₂ are particularly preferred for use in blowing agents and as a propellant in accordance with the present invention, particularly in combination with HFC0-1233zd.

[0180] The relative amount of any of the above noted additional compounds, which are contemplated for use in certain embodiments as co-blowing agents, as well as any additional components which may be included in present compositions, can vary widely within the general broad scope of the present invention according to the particular application for the composition, and all such relative amounts are considered to be within the scope hereof. Applicants note, however, that one particular advantage of at least certain of the compounds in accordance with the present invention is the relatively low flammability and relatively low toxicity of such compounds. Accordingly, in certain embodiments it is preferred that the composition of the present invention comprise at least one co-agent and an amount of the one or more monochlorotrifluoropropene compound(s) sufficient to produce a composition which is overall nonflammable. As used herein, the term "co-agent" is intended to refer to any one or more compounds which are included in the compositions for the purpose of contributing to at least some aspect of the performance of the composition for the intended purpose. Thus, in such embodiments, the relative amounts of the co-agent in comparison to the one or more monochlorotrifluoropropene compound(s) will depend, at least in part, upon the desirable properties of the composition, such as the flammability of the co-

agent.

[0181] The compositions of the present invention may include the compounds of the present invention in widely ranging amounts. It is generally preferred, however, that for preferred compositions for use as blowing agents in accordance with the present invention, monochlorotrifluoropropene compound(s) are present in an amount that is at least about 1% by weight, more preferably at least about 5% by weight, and even more preferably at least about 15% by weight, of the composition. In certain preferred embodiments, the blowing agent comprises at least about 50% by weight of the present blowing agent compound(s), and in certain embodiments the blowing agent consists essentially of compounds in accordance with the present invention. In this regard it is noted that the use of one or more co-blowing agents is consistent with the novel and basic features of the present invention. For example, it is contemplated that water will be used as either a co-blowing or in combination with other co-blowing agents (such as, for example, pentane, particularly cyclopentane) in a large number of embodiments.

[0182] It is contemplated that the blowing agent compositions of the present invention may comprise, preferably in amounts of at least about 15% by weight of the composition of one or more monochlorotrifluoropropene compound(s). In many preferred embodiments, a co-blowing agent comprising water is included in the compositions, most preferably in compositions directed to the use of thermosetting foams.

[0183] In certain embodiments, it is preferred that the blowing agent compositions of the present invention comprise HFC0-1233zd, more preferably at least about 90% by weight HFC0-1233zd, more preferably at least about 95% by weight HFC0-1233zd, and even more preferably at least about 99% by weight HFC0-1233zd. In certain preferred embodiments, it is preferred that the blowing agent compositions of the present invention comprise at least about 80%, and even more preferably at least about 90% by weight of HFCO-1233zd, and even more preferably any one or more of cis-HFCO-1233zd and trans-HFC-1233zd.

[0184] The blowing agent compositions of the present invention comprise in certain embodiments a combination of cisHFCO-1233zd and transHFCO1233zd. In certain embodiments, the cis:trans weight ratio is from about 30:70 to about 5:95, and even more preferably from about 20:80 to about 5:95, with a ration of 10:90 being especially preferred in certain embodiments.

[0185] In certain preferred embodiments, the blowing agent composition comprises from about 30% to about 95% by weight, more preferably from about 30% to about 96%, more preferably from about 30% to about 97%, and even more preferably from about 30% to about 98% by weight, and even more preferably from about 30% to about 99% by weight of one or more monochlorotrifluoropropene compound(s), and from about 5% to about 90% by weight, more preferably from about 5% to about 65% by weight of co-blowing agent, including one or more fluoroethers. In certain of such embodiments the co-blowing agent comprises, and preferably consists essentially of a compound selected from the group consisting of, H.sub.2O, HCs, HEs, HFCs, HFEs, hydrocarbons, alcohols (preferably C2, C3 and/or C4 alcohols), ketones, CO2, and combinations of any two or more of these.

[0186] In other embodiments, the invention provides foamable compositions. The foamable compositions of the present invention generally include one or more components capable of forming foam having. In certain embodiments, the one or more components comprise a thermosetting composition capable of forming foam and/or foamable compositions. Examples of thermosetting compositions include polyurethane and polyisocyanurate foam compositions, and also phenolic foam compositions. With respect to foam types, particularly polyurethane foam compositions, the present invention provides rigid foam (both closed cell, open cell and any combination thereof), flexible foam, and semiflexible foam, including integral skin foams. The present invention provides also single component foams, which include sprayable single component foams.

[0187] The reaction and foaming process may be enhanced through the use of various additives

such as catalysts and surfactant materials that serve to control and adjust cell size and to stabilize the foam structure during formation. Furthermore, it is contemplated that any one or more of the additional components described above with respect to the blowing agent compositions of the present invention could be incorporated into the foamable composition of the present invention. In such thermosetting foam embodiments, one or more of the present compositions are included as or part of a blowing agent in a foamable composition, or as a part of a two or more part foamable composition, which preferably includes one or more of the components capable of reacting and/or foaming under the proper conditions to form a foam or cellular structure.

[0188] In certain other embodiments, the one or more components comprise thermoplastic materials, particularly thermoplastic polymers and/or resins. Examples of thermoplastic foam components include polyolefins, such as for example monovinyl aromatic compounds of the formula $\text{Ar}-\text{CHCH}_2$ wherein Ar is an aromatic hydrocarbon radical of the benzene series such as polystyrene (PS), (PS). Other examples of suitable polyolefin resins in accordance with the invention include the various ethylene resins including the ethylene homopolymers such as polyethylene (PE), and ethylene copolymers, polypropylene (PP) and polyethyleneterephthalate (PET), and foams formed there from, preferably low-density foams. In certain embodiments, the thermoplastic foamable composition is an extrudable composition.

[0189] The invention also relates to foam, and preferably closed cell foam, prepared from a polymer foam formulation containing a blowing agent comprising the compositions of the invention. In yet other embodiments, the invention provides foamable compositions comprising thermoplastic or polyolefin foams, such as polystyrene (PS), polyethylene (PE), polypropylene (PP) and polyethyleneterephthalate (PET) foams, preferably low-density foams.

D. Compositions Containing Trifluoromonochloropropene

[0190] Applicants have developed several compositions which include as an essential component one or more trifluoromonochloropropene, compounds, including $\text{transCF}_3\text{CH}=\text{CClH}$ (1233zdE), $\text{cisCF}_3\text{CH}=\text{CClH}$ (1233zdZ), $\text{transCHF}_2\text{CF}=\text{CClH}$ (1233ydE), $\text{cisCHF}_2\text{CF}=\text{CClH}$ (1233ydZ), $\text{transCHF}_2\text{CH}=\text{CClF}$ (1233zbE), $\text{cisCHF}_2\text{CH}=\text{CClF}$ (1233zbZ), $\text{transCHF}_2\text{CCl}=\text{CHF}$ (1233xeE), $\text{cisCHF}_2\text{CCl}=\text{CHF}$ (1233xeZ), $\text{CH}_2\text{FCCl}=\text{CF}_2$ (1233xc), $\text{transCHFC1CF}=\text{CFH}$ (1233yeE), $\text{cisCHFC1CF}=\text{CFH}$ (1233yeZ), $\text{CH}_2\text{C1CF}=\text{CF}_2$ (1233yc), including all combinations of these in all proportions, and at least one additional compound. In such compositions, the amount of the one or more trifluoromonochloropropene(s) may vary widely, including in all cases constituting the balance of the composition after all other components in composition are accounted for. In certain preferred embodiments, the amount of each of the above noted trifluoromonochloropropene, and the amount of any combination of two or more of these in any and all proportions, in the composition can be in accordance with the following ranges: from about 1 wt % to about 99 wt %; from about 80 wt % to about 99 wt %; from about 1 wt % to about 20 wt %; from about 1 wt % to about 25 wt %; from about 1 wt % to about 30 wt %; and from about 1 wt % to about 50 wt %. Preferred compositions of this type are described in the Table below, with all percentages being in percent by weight and being understood to be preceded by the word "about" in connection with the additional compound specified in the table. In addition, it will be understood that the Table below applies to each of $\text{transCF}_3\text{CH}=\text{CClH}$ (1233zdE), $\text{cisCF}_3\text{CH}=\text{CClH}$ (1233zdZ), $\text{transCHF}_2\text{CF}=\text{CClH}$ (1233ydE), $\text{cisCHF}_2\text{CF}=\text{CClH}$ (1233ydZ), $\text{transCHF}_2\text{CH}=\text{CClF}$ (1233zbE), $\text{cisCHF}_2\text{CH}=\text{CClF}$ (1233zbZ), $\text{transCHF}_2\text{CCl}=\text{CHF}$ (1233xeE), $\text{cisCHF}_2\text{CCl}=\text{CHF}$ (1233xeZ), $\text{CH}_2\text{FCCl}=\text{CF}_2$ (1233xc), $\text{transCHFC1CF}=\text{CFH}$ (1233yeE), $\text{cisCHFC1CF}=\text{CFH}$ (1233yeZ), $\text{CH}_2\text{C1CF}=\text{CF}_2$ (1233yc) and all combinations and proportions of these two compounds.

TABLE-US-00001 COMBINATIONS WITH HFCO-1233* Preferred Second Third

ADDITIONAL COMPOUND	Range	Preferred	Preferred	HFOs	Chlorofluorobutenes	1 to 99
1 to 99	1 to 20	80 to 99	Tetrafluorobutene	1 to 99	1 to 20	80 to 99
Pentafluorobutene	1 to 99	1 to 20	80 to 99	HFO-1354	1 to 99	1 to 20
80 to 99	HFO-1345	1 to 99	1 to 20	80 to 99	trans-HFO-1234ze	1 to 99

1 to 20 80 to 99 cis-HFO1234ze 1 to 99 1 to 20 80 to 99 HFO-1234yf 1 to 99 1 to 20 80 to 99 HF01225ye 1 to 99 1 to 50 na CF3CH—CHCF3 (E & Z) 1 to 99 1 to 50 na (CF3)2CFCH=CHF (E & Z) 1 to 99 1 to 50 na (CF3)2CFCH=CF2 1 to 99 1 to 50 na CF3CHFC—CHF (E & Z) 1 to 99 1 to 50 na (C2F5)(CF3)C=CH2 1 to 99 1 to 50 na Trifluoropropene (all isomers) 1 to 99 1 to 50 na (C2F5)(CF3)C=CH2 1 to 99 1 to 50 na transCHF=CFCHF2 (HFO-I 234yeZ) 1 to 99 1 to 20 80 to 99 cisCHF=CFCHF2 (HFO-1234yeE) 1 to 99 1 to 20 80 to 99 HFCs HFC-245fa 1 to 99 1 to 70 na HFC-245eb 1 to 99 1 to 70 na HFC-245ca 1 to 99 1 to 70 na HFC-227ea 1 to 99 1 to 70 na HFC-236ea 1 to 99 1 to 70 na HFC-236fa 1 to 99 1 to 70 na HFC-134a 1 to 99 1 to 70 na HFC-134 I to 99 1 to 70 na HFC-152a 1 to 99 1 to 70 na HFC-32 1 to 99 1 to 70 na HFC-125 1 to 99 1 to 70 na HFC-143a 1 to 99 1 to 70 na HFC-365mfc 1 to 99 1 to 70 na HFC-161 1 to 99 1 to 70 na HFC-43-10mee 1 to 99 1 to 70 na HFEs CHF2—O—CHF2 1 to 99 na na CHF2—O—CH2F 1 to 99 na na CH2F—O—CH2F 1 to 99 na na CH2F—O—CH3 1 to 99 na na CYCLO—CF2—CH2—CF2—O 1 to 99 na na CYCLO—CF2—CF2—CH2—O 1 to 99 na na CHF2—O—CF2—CHF2 1 to 99 na na CF3—CF2—O—CH2F 1 to 99 na na CHF2—O—CHF—CF3 1 to 99 na na CHF2—O—CF2—CHF2 1 to 99 na na CH2F—O—CF2—CHF2 1 to 99 na na CF3—O—CF2—CH3 1 to 99 na na CHF2—CHF—O—CHF2 1 to 99 na na CF3—O—CHF—CH2F 1 to 99 na na CF3—CHF—O—CH2F 1 to 99 na na CF3—O—CH2—CHF2 1 to 99 na na CHF2—O—CH2—CF3 1 to 99 na na CH2F—CF2—O—CH2F 1 to 99 na na CHF2—O—CF2—CH3 1 to 99 na na CHF2—CF2—O—CH3 (254pc) 1 to 99 na na CH2F—O—CHF—CH2F 1 to 99 na na CHF2—CHF—O—CH2F 1 to 99 na na CF3—O—CHF—CH3 1 to 99 na na CF3—CHF—O—CH3 1 to 99 na na CHF2—O—CH2—CHF2 1 to 99 na na CF3—O—CH2—CH2F 1 to 99 na na CF3—CH2—O—CH2F 1 to 99 na na CF2H—CF2—CF2—O—CH3 1 to 99 na na Hydrocarbons 1 to 99 na Propane 1 to 99 na na Butane 1 to 99 na na Isobutane 1 to 99 na na n-pentane (high HFO) 1 to 99 na na n-pentane (high n-pentane) 1 to 99 na na Isopentane (High HFO) 1 to 99 na na Isopentane (High 1 to 99 na na Neopentane (High HFO) 1 to 99 na na Neopentane (High neopentane) 1 to 99 na na Cyclopentane (High HFO) 1 to 99 na na Cyclopentane (High cyclopentane) 1 to 99 na na n-hexane 1 to 99 na na Isohexane 1 to 99 na na Heptane 1 to 99 na na Alcohol Methanol 1 to 50 1 to 20 na Ethanol 1 to 50 1 to 20 na Propanal 1 to 50 1 to 20 na Isopropanol 1 to 50 1 to 20 na Butanol 1 to 50 1 to 20 na Isobutanol 1 to 50 1 to 20 na t-butanol 1 to 50 1 to 20 na Ethers Dimethylether 1 to 50 na na Methylethylether 1 to 50 1 to 30 na diethyl ether 1 to 50 Na Na Methylpropylether 1 to 50 1 to 30 na methylisopropylether 1 to 50 1 to 30 na Ethylpropylether 1 to 50 1 to 30 na Ethylisopropylether 1 to 50 1 to 30 na Dipropylether 1 to 50 1 to 30 na Diisopropylether 1 to 50 1 to 30 na dimethyloxymethane 1 to 50 1 to 30 na Diethoxymethane 1 to 50 1 to 30 na Dipropoxymethane 1 to 50 1 to 30 na Dibutoxymethane 1 to 50 1 to 30 na Aldehydes Formaldehyde 1 to 99 10 to 90 10 to 80 Acetaldehyde 1 to 99 10 to 90 10 to 80 Propanal 1 to 99 10 to 90 10 to 80 Butanal 1 to 99 10 to 90 10 to 80 Isobutanal 1 to 99 10 to 90 10 to 80 Ketones Acetone 1 to 50 1 to 30 na Methylethylketone 1 to 50 1 to 30 na Methylisobutylketone 1 to 50 1 to 30 na Others methyl formate 1 to 99 10 to 90 10 to 80 formic acid 1 to 99 10 to 90 10 to 80 Water 1 to 99 1 to 50 1 to 30 trans-1,2 dichloroethylene 1 to 99 1 to 50 1 to 30 Carbon dioxide 1 to 99 10 to 90 10 to 80 cis-HFO-1234ze + HFO- 1225yez 1 to 25/1 Na Na to 50 All the above plus water All the above plus CO2 All the above plus trans 1,2- All the above plus methyl formate cis-HFO-1234ze + CO2 cis-HFO-1234ze + HFO-1225yez + cis-HFO-1234ze + HFC-245fa 1 to 25/1 1 to 20/1 1 to 15/1 *Each combination is understood to be with each of the above-noted compounds, and all combinations of each with the others.

[0191] In preferred embodiments in which the co-agent comprises H.sub.2O, the composition comprises H.sub.2O in an amount of from about 5% by weight to about 50% by weight of the total composition, more preferably from about 10% by weight to about 40% by weight, and even more preferably of from about 10% to about 20% by weight of the total composition.

[0192] In preferred embodiments in which the co-agent comprises CO.sub.2, the composition comprises CO.sub.2 in an amount of from about 5% by weight to about 60% by weight of the total

composition, more preferably from about 20% by weight to about 50% by weight, and even more preferably of from about 40% to about 50% by weight of the composition.

[0193] In preferred embodiments in which the co-agent comprises alcohols, (preferably C2, C3 and/or C4 alcohols), the composition comprises alcohol in an amount of from about 5% by weight to about 40% by weight of the total composition, more preferably from about 10% by weight to about 40% by weight, and even more preferably of from about 15% to about 25% by weight of the total composition.

[0194] For compositions which include HFC co-agents, the HFC co-blowing agent (preferably C2, C3, C4 and/or C5 HFC), and even more preferably difluoromethane (HFC-152a) (HFC-152a being particularly preferred for compositions uses as blowing agents for extruded thermoplastics) and/or pentafluoropropane (HFC-245)), is preferably present in the composition in amounts of from of from about 5% by weight to about 80% by weight of the composition, more preferably from about 10% by weight to about 75% by weight, and even more preferably of from about 25% to about 75% by weight of the composition. Furthermore, in such embodiments, the HFC is preferably C2-C4 HFC, and even more preferably C3 HFC, with penta-fluorinated C3 HFC, such as HFC-245fa, being highly preferred in certain embodiments.

[0195] For compositions which include HFE co-agents, the HFE co-agent (preferably C2, C3, C4 and/or C5 HFE), and even more preferably HFE-254 (including particularly HFE-254pc) is preferably present in the composition in amounts of from of from about 5% by weight to about 80% by weight of the total composition, more preferably from about 10% by weight to about 75% by weight, and even more preferably of from about 25% to about 75% by weight of the composition. Furthermore, in such embodiments, the HFE is preferably C2-C4 HFE, and even more preferably a C3 HFC, with tetra-fluorinated C3 HFE being highly preferred in certain embodiments.

[0196] For compositions which include HC co-agents, the HC co-agent (preferably C3, C4 and/or C5 HC) is preferably present in the composition in amounts of from of from about 5% by weight to about 80% by weight of the total composition, and even more preferably from about 20% by weight to about 60% by weight of the composition.

E. Methods and Systems

1. Foam Forming Methods

[0197] It is contemplated that all presently known and available methods and systems for forming foam are readily adaptable for use in connection with the present invention. For example, the methods of the present invention generally require incorporating a blowing agent in accordance with the present invention into a foamable or foam forming composition and then foaming the composition, preferably by a step or series of steps which include causing volumetric expansion of the blowing agent in accordance with the present invention. In general, it is contemplated that the presently used systems and devices for incorporation of blowing agent and for foaming are readily adaptable for use in accordance with the present invention. In fact, it is believed that one advantage of the present invention is the provision of an improved blowing agent which is generally compatible with existing foaming methods and systems.

[0198] Thus, it will be appreciated by those skilled in the art that the present invention comprises methods and systems for foaming all types of foams, including thermosetting foams, thermoplastic foams and formed-in-place foams. Thus, one aspect of the present invention is the use of the present blowing agents in connection conventional foaming equipment, such as polyurethane foaming equipment, at conventional processing conditions. The present methods therefore include masterbatch type operations, blending type operations, third stream blowing agent addition, and blowing agent addition at the foam head.

[0199] With respect to thermoplastic foams, the preferred methods generally comprise introducing a blowing agent in accordance with the present invention into a thermoplastic material, preferably thermoplastic polymer such as polyolefin, and then subjecting the thermoplastic material to

conditions effective to cause foaming. For example, the step of introducing the blowing agent into the thermoplastic material may comprise introducing the blowing agent into a screw extruder containing the thermoplastic, and the step of causing foaming may comprise lowering the pressure on the thermoplastic material and thereby causing expansion of the blowing agent and contributing to the foaming of the material.

[0200] It will be appreciated by those skilled in the art, especially in view of the disclosure contained herein, that the order and manner in which the blowing agent of the present invention is formed and/or added to the foamable composition does not generally affect the operability of the present invention. For example, in the case of extrudable foams, it is possible that the various components of the blowing agent, and even the components of the present composition, be not be mixed in advance of introduction to the extrusion equipment, or even that the components are not added to the same location in the extrusion equipment. Moreover, the blowing agent can be introduced either directly or as part of a premix, which is then further added to other parts of the foamable composition.

[0201] Thus, in certain embodiments it may be desired to introduce one or more components of the blowing agent at first location in the extruder, which is upstream of the place of addition of one or more other components of the blowing agent, with the expectation that the components will come together in the extruder and/or operate more effectively in this manner. Nevertheless, in certain embodiments, two or more components of the blowing agent are combined in advance and introduced together into the foamable composition, either directly or as part of premix which is then further added to other parts of the foamable composition.

[0202] In certain preferred embodiments, dispersing agents, cell stabilizers, surfactants and other additives may also be incorporated into the blowing agent compositions of the present invention. Surfactants are optionally but preferably added to serve as cell stabilizers. Some representative materials are sold under the names of DC-193, B-8404, and L-5340 which are, generally, polysiloxane polyoxyalkylene block co-polymers such as those disclosed in U.S. Pat. Nos. 2,834,748, 2,917,480, and 2,846,458, each of which is incorporated herein by reference. Other optional additives for the blowing agent mixture may include flame retardants such as tri(2-chloroethyl)phosphate, tri(2-chloropropyl)phosphate, tri(2,3-dibromopropyl)-phosphate, tri(1,3-dichloropropyl) phosphate, diammonium phosphate, various halogenated aromatic compounds, antimony oxide, aluminum trihydrate, polyvinyl chloride, and the like.

[0203] Any of the methods well known in the art, such as those described in "Polyurethanes Chemistry and Technology," Volumes I and II, Saunders and Frisch, 1962, John Wiley and Sons, New York, NY, which is incorporated herein by reference, may be used or adapted for use in accordance with the foam embodiments of the present invention.

2. Propellant and Aerosol Compositions

[0204] In another aspect, the present invention provides propellant compositions comprising or consisting essentially of a composition of the present invention. In certain preferred embodiments, such propellant composition is preferably a sprayable composition, either alone or in combination with other known propellants.

[0205] In one aspect, the present compositions may be used for propelling objects, including solid and/or liquid objects and/or gaseous objects, by applying to such objects a force generated by the present composition, such as would occur through the expansion of the compositions of the present invention. For example, such force may preferably be provided, at least in part, by the change of phase of the compositions of the present invention from liquid to gas, and/or by the force released as a result of a substantial pressure reduction as the composition of the present invention exits from a pressurized container. In this way, the compositions of the present invention may be used to apply a burst of force, or a sustained force to an object to be propelled. Accordingly, the present invention comprises systems, containers and devices which include compositions of the present invention and which are configured to propel or move an object, either a liquid object or a solid object or a

gaseous object, with the desired amount of force. Examples of such uses include containers (such as pressurized cans and similar devices) which may be used, through the propellant force, to unblock drains, pipes or blockages in conduits, channels or nozzles. Another application includes use of the present composition to propel solid objects through the environment, particularly the ambient air, such as bullets, pellets, grenades, nets, canisters, bean bags, electrodes or other individual tethered or untethered projectiles. In other embodiments, the present compositions may be used to impart motion, such as a spitting motion, to gyroscopes, centrifuges, toys or other bodies to be rotated, or to impart a propelling force to solid objects, such as fireworks, confetti, pellets, munitions and other solid objects. In other applications, the force provided by the compositions of the present invention may be used to push or steer bodies in motion, including rockets or other projectiles.

[0206] The propellant compositions of the present invention preferably comprise a material to be sprayed and a propellant comprising, consisting essentially of, or consisting of a composition in accordance with the present invention. Inert ingredients, solvents, and other materials may also be present in the sprayable mixture. Preferably, the sprayable composition is an aerosol. Suitable materials to be sprayed include, without limitation, cosmetic materials such as deodorants, perfumes, hair sprays, cleaning solvents, and lubricants, as well as medicinal materials such as anti-asthma medications. The term medicinal materials is used herein in its broadest sense to include any and all materials which are, or at least are believed to be, effective in connection with therapeutic treatments, diagnostic methods, pain relief, and similar treatments, and as such would include for example drugs and biologically active substances. The medicinal material in certain preferred embodiments are adapted to be inhaled. The medicament or other therapeutic agent is preferably present in the composition in a therapeutic amount, with a substantial portion of the balance of the composition comprising a one or more monochlorotrifluoropropene compound(s) of the present invention, as described above.

[0207] Aerosol products for industrial, consumer or medical use typically contain one or more propellants along with one or more active ingredients, inert ingredients or solvents. The propellant provides the force that expels the product in aerosolized form. While some aerosol products are propelled with compressed gases like carbon dioxide, nitrogen, nitrous oxide and even air, most commercial aerosols use liquefied gas propellants. The most commonly used liquefied gas propellants are hydrocarbons such as butane, isobutane, and propane. Dimethyl ether and HFC-152a (1,1-difluoroethane) are also used, either alone or in blends with the hydrocarbon propellants. Unfortunately, all of these liquefied gas propellants are highly flammable and their incorporation into aerosol formulations will often result in flammable aerosol products.

[0208] Applicants have come to appreciate the continuing need for nonflammable, liquefied gas propellants with which to formulate aerosol products. The present invention provides compositions of the present invention, particularly and preferably compositions comprising HFCO-1233 as described above, for use in certain industrial aerosol products, including for example spray cleaners, lubricants, and the like, and in medicinal aerosols, including for example to deliver medications to the lungs or mucosal membranes. Examples of this includes metered dose inhalers (MDIs) for the treatment of asthma and other chronic obstructive pulmonary diseases and for delivery of medicaments to accessible mucous membranes or intranasally. The present invention thus includes methods for treating ailments, diseases and similar health related problems of an organism (such as a human or animal) comprising applying a composition of the present invention containing a medicament or other therapeutic component to the organism in need of treatment. In certain preferred embodiments, the step of applying the present composition comprises providing a MDI containing the composition of the present invention (for example, introducing the composition into the MDI) and then discharging the present composition from the MDI.

[0209] The compositions of the present invention, particularly compositions which comprise or consist essentially of any one or more of monochlorotrifluoropropenes of the present invention, are

capable of providing nonflammable, liquefied gas propellant and aerosols that do not contribute substantially to global warming. The present compositions can be used to formulate a variety of industrial aerosols or other sprayable compositions such as contact cleaners, dusters, lubricant sprays, and the like, and consumer aerosols such as personal care products, household products and automotive products. The medicinal aerosol and/or propellant and/or sprayable compositions of the present invention in many applications include, in addition to compounds of the present invention, a medicament such as a beta-agonist, a corticosteroid or other medicament, and, optionally, other ingredients, such as surfactants, solvents, other propellants, flavorants and other excipients. The compositions of the present invention, unlike many compositions previously used in these applications, have good environmental properties and are not considered to be potential contributors to global warming. The present compositions therefore provide in certain preferred embodiments substantially nonflammable, liquefied gas propellants having very low Global Warming potentials.

3. Flavorants and Fragrances

[0210] The compositions of the present invention also provide advantage when used as part of, and in particular as a carrier for, flavor formulations and fragrance formulations. The suitability of the present compositions for this purpose is demonstrated by a test procedure in which a predetermined amount of a plant material, such as Jasnone, are put into a heavy walled glass tube and an amount of one or more of the compounds of the present invention are added to the glass tube. The tube are then frozen and sealed. Upon thawing the tube, it is found that the mixture had one liquid phase, thus establishing favorable use of the one or more monochlorotetrafluoropropenes as a carrier for flavor formulations and fragrances. It also establishes its potential as an extractant of biologically active compounds (such as Biomass) and fragrances, including from plant matter. In certain embodiments, it may be preferred to use the present composition for in extraction applications with the present fluid in its supercritical state. This and other applications of involving use of the present compositions in the supercritical or near supercritical state are described hereinafter.

4. Inflating Agents

[0211] One potential advantage of the compositions of the present invention is that the preferred compositions are in a gaseous state under most ambient conditions. This characteristic allows them to fill the space while not adding significantly to the weight of the space being spilled. Furthermore, the compositions of the present invention are able to be compressed or liquefied for relatively easy transportation and storage. Thus, for example, the compositions of the present invention may be included, preferably but not necessarily in liquid form, in a closed container, such as a pressurized can, which has a nozzle therein adapted to release the composition into another environment in which it will exist, at least for a period of time, as a pressurized gas. For example, such an application may include including the present compositions in a can adapted to connect to tires such as may be used on transportation vehicles (including cars, trucks and aircraft). Other examples in accordance with this embodiment include the use of the present compositions, in a similar arrangement, to inflate air bags or other bladders (including other protective bladders) adapted to contain, at least for a period of time, a gaseous material under pressure. Alternatively to the use of a fixed container, such as I can, the present compositions may be applied in accordance with this aspect of the invention through a hose or other system that contains the present composition, either in liquid or gaseous form, and through which it can be introduced into such a pressurized environment as is required for the particular application.

F. Methods and Systems

[0212] The compositions of the present invention are useful in connection with numerous methods and systems, including as heat transfer fluids in methods and systems for transferring heat, such as refrigerants used in refrigeration, air conditioning and heat pump systems. The present compositions are also advantageous for in use in systems and methods of generating aerosols, preferably comprising or consisting of the aerosol propellant in such systems and methods.

Methods of forming foams and methods of extinguishing and suppressing fire are also included in certain aspects of the present invention. The present invention also provides in certain aspects methods of removing residue from articles in which the present compositions are used as solvent compositions in such methods and systems.

1. Heat Transfer Methods and Systems

[0213] The preferred heat transfer methods generally comprise providing a composition of the present invention and causing heat to be transferred to or from the composition, either by sensible heat transfer, phase change heat transfer, or a combination of these. For example, in certain preferred embodiments the present methods provide refrigeration systems comprising a refrigerant of the present invention and methods of producing heating or cooling by condensing and/or evaporating a composition of the present invention. In certain preferred embodiments, the methods for cooling, including cooling of other fluid either directly or indirectly or a body directly or indirectly, comprise condensing a refrigerant composition comprising a composition of the present invention and thereafter evaporating said refrigerant composition in the vicinity of the article to be cooled. As used herein, the term “body” is intended to refer not only to inanimate objects but also to living tissue, including animal tissue in general and human tissue in particular. For example, certain aspects of the present invention involve application of the present composition to human tissue for one or more therapeutic purposes, such as a pain killing technique, as a preparatory anesthetic, or as part of a therapy involving reducing the temperature of the body being treated. In certain embodiments, the application to the body comprises providing the present compositions in liquid form under pressure, preferably in a pressurized container having a one-way discharge valve and/or nozzle, and releasing the liquid from the pressurized container by spraying or otherwise applying the composition to the body. As the liquid evaporates from the surface being sprayed, the surface cools.

[0214] Certain preferred methods for heating a fluid or body comprise condensing a refrigerant composition comprising a composition of the present invention in the vicinity of the fluid or body to be heated and thereafter evaporating said refrigerant composition. In light of the disclosure herein, those of skill in the art will be readily able to heat and cool articles according to the present inventions without undue experimentation.

[0215] Applicants have found that in the systems and methods of the present invention many of the important refrigeration system performance parameters are relatively close to the parameters for R-134a. Since many existing refrigeration systems have been designed for R-134a, or for other refrigerants with properties similar to R-134a, those skilled in the art will appreciate the substantial advantage of a low GWP and/or a low ozone depleting refrigerant that can be used as replacement for R-134a or like refrigerants with relatively minimal modifications to the system. It is contemplated that in certain embodiments the present invention provides retrofitting methods which comprise replacing the heat transfer fluid (such as a refrigerant) in an existing system with a composition of the present invention, without substantial modification of the system. In certain preferred embodiments the replacement step is a drop-in replacement in the sense that no substantial redesign of the system is required and no major item of equipment needs to be replaced in order to accommodate the composition of the present invention as the heat transfer fluid. In certain preferred embodiments, the methods comprise a drop-in replacement in which the capacity of the system is at least about 70%, preferably at least about 85%, and even more preferably at least about 90% of the system capacity prior to replacement. In certain preferred embodiments, the methods comprise a drop-in replacement in which the suction pressure and/or the discharge pressure of the system, and even more preferably both, is/are at least about 70%, more preferably at least about 90% and even more preferably at least about 95% of the suction pressure and/or the discharge pressure prior to replacement. In certain preferred embodiments, the methods comprise a drop-in replacement in which the mass flow of the system is at least about 80%, and even more preferably at least 90% of the mass flow prior to replacement.

[0216] In certain embodiments the present invention provides cooling by absorbing heat from a fluid or body, preferably by evaporating the present refrigerant composition in the vicinity of the body or fluid to be cooled to produce vapor comprising the present composition. Preferably the methods include the further step of compressing the refrigerant vapor, usually with a compressor or similar equipment to produce vapor of the present composition at a relatively elevated pressure. Generally, the step of compressing the vapor results in the addition of heat to the vapor, thus causing an increase in the temperature of the relatively high pressure vapor. Preferably in such embodiments the present methods include removing from this relatively high temperature, high pressure vapor at least a portion of the heat added by the evaporation and compression steps. The heat removal step preferably includes condensing the high temperature, high pressure vapor while the vapor is in a relatively high pressure condition to produce a relatively high pressure liquid comprising a composition of the present invention. This relatively high pressure liquid preferably then undergoes a nominally isoenthalpic reduction in pressure to produce a relatively low temperature, low pressure liquid. In such embodiments, it is this reduced temperature refrigerant liquid which is then vaporized by heat transferred from the body or fluid to be cooled.

[0217] In another process embodiment of the invention, the compositions of the invention may be used in a method for producing heating which comprises condensing a refrigerant comprising the compositions in the vicinity of a liquid or body to be heated. Such methods, as mentioned hereinbefore, frequently are reverse cycles to the refrigeration cycle described above.

2. Foam Blowing Methods

[0218] One embodiment of the present invention relates to methods of forming foams, and preferably polyurethane and polyisocyanurate foams. The methods generally comprise providing a blowing agent composition of the present inventions, adding (directly or indirectly) the blowing agent composition to a foamable composition, and reacting the foamable composition under the conditions effective to form a foam or cellular structure, as is well known in the art. Any of the methods well known in the art, such as those described in "Polyurethanes Chemistry and Technology," Volumes I and II, Saunders and Frisch, 1962, John Wiley and Sons, New York, NY, which is incorporated herein by reference, may be used or adapted for use in accordance with the foam embodiments of the present invention. In general, such preferred methods comprise preparing polyurethane or polyisocyanurate foams by combining an isocyanate, a polyol or mixture of polyols, a blowing agent or mixture of blowing agents comprising one or more of the present compositions, and other materials such as catalysts, surfactants, and optionally, flame retardants, colorants, or other additives.

[0219] It is convenient in many applications to provide the components for polyurethane or polyisocyanurate foams in pre-blended formulations. Most typically, the foam formulation is pre-blended into two components. The isocyanate and optionally certain surfactants and blowing agents comprise the first component, commonly referred to as the "A" component. The polyol or polyol mixture, surfactant, catalysts, blowing agents, flame retardant, and other isocyanate reactive components comprise the second component, commonly referred to as the "B" component.

Accordingly, polyurethane or polyisocyanurate foams are readily prepared by bringing together the A and B side components either by hand mix for small preparations and, preferably, machine mix techniques to form blocks, slabs, laminates, pour-in-place panels and other items, spray applied foams, froths, and the like. Optionally, other ingredients such as fire retardants, colorants, auxiliary blowing agents, and even other polyols can be added as a third stream to the mix head or reaction site. Most preferably, however, they are all incorporated into one B-component as described above.

[0220] It is also possible to produce thermoplastic foams using the compositions of the invention. For example, conventional polystyrene and polyethylene formulations may be combined with the compositions in a conventional manner to produce rigid foams.

3. Cleaning Methods

[0221] The present invention also provides methods of removing containments from a product,

part, component, substrate, or any other article or portion thereof by applying to the article a composition of the present invention. For the purposes of convenience, the term “article” is used herein to refer to all such products, parts, components, substrates, and the like and is further intended to refer to any surface or portion thereof. Furthermore, the term “contaminant” is intended to refer to any unwanted material or substance present on the article, even if such substance is placed on the article intentionally. For example, in the manufacture of semiconductor devices it is common to deposit a photoresist material onto a substrate to form a mask for the etching operation and to subsequently remove the photoresist material from the substrate. The term “contaminant” as used herein is intended to cover and encompass such a photo resist material.

[0222] In certain preferred methods, the cleaning step includes the step of flushing a material, such lubricants, from a vessel or container in connection with steps for preparing the system for retrofitting and/or regeneration. Such methods are in certain embodiments associated with retrofitting or replacing in an existing heat transfer system, such as a refrigeration or air conditioning system, the old refrigerant with a new refrigerant and flushing the system using a composition of the present invention as part of the process, particularly to remove at least a portion, and preferably substantially all, of previously used lubricant present in such systems

[0223] Preferred methods of the present invention comprise applying the present composition to the article. Although it is contemplated that numerous and varied cleaning techniques can employ the compositions of the present invention to good advantage, it is considered to be particularly advantageous to use the present compositions in connection with supercritical cleaning techniques. Supercritical cleaning is disclosed in U.S. Pat. No. 6,589,355, which is assigned to the assignee of the present invention and incorporated herein by reference. For supercritical cleaning applications, is preferred in certain embodiments to include in the present cleaning compositions, in addition to the HFC0-1233 one or more additional components, such as: HFO-1234 (preferably any one or more of cis-HFO-1234ze, trans-HFO-1234ze, HFO-1234yf, HFO-1234yc, HFO-1234zc, HFO-1234ye(E) and HFO-1234ye(Z)), CO₂ and other additional components known for use in connection with supercritical cleaning applications. It may also be possible and desirable in certain embodiments to use the present cleaning compositions in connection with particular vapor degreasing and solvent cleaning methods.

4. Flammability Reduction Methods

[0224] According to certain other preferred embodiments, the present invention provides methods for reducing the flammability of fluids, said methods comprising adding a compound or composition of the present invention to said fluid. The flammability associated with any of a wide range of otherwise flammable fluids may be reduced according to the present invention. For example, the flammability associated with fluids such as ethylene oxide, flammable hydrofluorocarbons and hydrocarbons, including: HFC-152a, 1,1,1-trifluoroethane (HFC-143a), difluoromethane (HFC-32), propane, hexane, octane, and the like can be reduced according to the present invention. For the purposes of the present invention, a flammable fluid may be any fluid exhibiting flammability ranges in air as measured via any standard conventional test method, such as ASTM E-681, and the like.

[0225] Any suitable amounts of the present compounds or compositions may be added to reduce flammability of a fluid according to the present invention. As will be recognized by those of skill in the art, the amount added will depend, at least in part, on the degree to which the subject fluid is flammable and the degree to which it is desired to reduce the flammability thereof. In certain preferred embodiments, the amount of compound or composition added to the flammable fluid is effective to render the resulting fluid substantially non-flammable.

5. Flame Suppression Methods

[0226] The present invention further provides methods of suppressing a flame, said methods comprising contacting a flame with a fluid comprising a compound or composition of the present invention. Any suitable methods for contacting the flame with the present composition may be

used. For example, a composition of the present invention may be sprayed, poured, and the like onto the flame, or at least a portion of the flame may be immersed in the composition. In light of the teachings herein, those of skill in the art will be readily able to adapt a variety of conventional apparatus and methods of flame suppression for use in the present invention.

6. Sterilization Methods

[0227] Many articles, devices and materials, particularly for use in the medical field, must be sterilized prior to use for the health and safety reasons, such as the health and safety of patients and hospital staff. The present invention provides methods of sterilizing comprising contacting the articles, devices or material to be sterilized with a compound or composition of the present invention comprising one or more of the HFCO-1233 compounds described herein, in combination with one or more sterilizing agents. While many sterilizing agents are known in the art and are considered to be adaptable for use in connection with the present invention, in certain preferred embodiments sterilizing agent comprises ethylene oxide, formaldehyde, hydrogen peroxide, chlorine dioxide, ozone and combinations of these. In certain embodiments, ethylene oxide is the preferred sterilizing agent. Those skilled in the art, in view of the teachings contained herein, will be able to readily determine the relative proportions of sterilizing agent and the present compound(s) to be used in connection with the present sterilizing compositions and methods, and all such ranges are within the broad scope hereof. As is known to those skilled in the art, certain sterilizing agents, such as ethylene oxide, are relatively flammable components, and the compound(s) in accordance with the present invention are included in the present compositions in amounts effective, together with other components present in the composition, to reduce the flammability of the sterilizing composition to acceptable levels.

[0228] The sterilization methods of the present invention may be either high or low-temperature sterilization of the present invention involves the use of a compound or composition of the present invention at a temperature of from about 250° F. to about 270° F., preferably in a substantially sealed chamber. The process can be completed usually in less than about 2 hours. However, some articles, such as plastic articles and electrical components, cannot withstand such high temperatures and require low-temperature sterilization. In low temperature sterilization methods, the article to be sterilized is exposed to a fluid comprising a composition of the present invention at a temperature of from about room temperature to about 200° F., more preferably at a temperature of from about room temperature to about 100° F.

[0229] The low-temperature sterilization of the present invention is preferably at least a two-step process performed in a substantially sealed, preferably air tight, chamber. In the first step (the sterilization step), the articles having been cleaned and wrapped in gas permeable bags are placed in the chamber. Air is then evacuated from the chamber by pulling a vacuum and perhaps by displacing the air with steam. In certain embodiments, it is preferable to inject steam into the chamber to achieve a relative humidity that ranges preferably from about 30% to about 70%. Such humidities may maximize the sterilizing effectiveness of the sterilant which is introduced into the chamber after the desired relative humidity is achieved. After a period of time sufficient for the sterilant to permeate the wrapping and reach the interstices of the article, the sterilant and steam are evacuated from the chamber.

[0230] In the preferred second step of the process (the aeration step), the articles are aerated to remove sterilant residues. Removing such residues is particularly important in the case of toxic sterilants, although it is optional in those cases in which the substantially nontoxic compounds of the present invention are used. Typical aeration processes include air washes, continuous aeration, and a combination of the two. An air wash is a batch process and usually comprises evacuating the chamber for a relatively short period, for example, 12 minutes, and then introducing air at atmospheric pressure or higher into the chamber. This cycle is repeated any number of times until the desired removal of sterilant is achieved. Continuous aeration typically involves introducing air through an inlet at one side of the chamber and then drawing it out through an outlet on the other

side of the chamber by applying a slight vacuum to the outlet. Frequently, the two approaches are combined. For example, a common approach involves performing air washes and then an aeration cycle.

7. Supercritical Methods

[0231] It is contemplated that in general many of the uses and methods described herein can be carried out with the present compositions in the supercritical or near supercritical state. For example, the present compositions may be utilized in solvent and solvent extraction applications mentioned herein, particularly for use in connection with materials such as alkaloids (which are commonly derived from plant sources), for example caffeine, codeine and papaverine, for organometallic materials such as metallocenes, which are generally useful as catalysts, and for fragrances and flavors such as Jasmone.

[0232] The present compositions, preferably in their supercritical or near supercritical state, can be used in connection with methods involving the deposit of catalysts, particularly organometallic catalysts, on solid supports. In one preferred embodiment, these methods include the step of generating finely divided catalyst particles, preferably by precipitating such catalyst particles from the present compositions in the supercritical or near supercritical state. It is expected that in certain preferred embodiments catalysts prepared in accordance with the present methods will exhibit excellent activity.

[0233] It is also contemplated that certain of the MDI methods and devices described herein may utilize medicaments in finely divided form, and in such situations it is contemplated that the present invention provides methods which include the step of incorporating such finely divided medicament particles, such as albuterol, into the present fluids, preferably by dissolving such particles, in the present composition, preferably in the supercritical or near supercritical state. In cases where the solubility of the materials is relatively low when the present fluids are in the supercritical or near supercritical state, it may be preferred to use entrainers such as alcohols.

[0234] It is also contemplated that the present compositions in the supercritical or near supercritical state may be used to clean circuit boards and other electronic materials and articles.

[0235] Certain materials may have very limited solubility in the present compositions, particularly when in the supercritical or near supercritical state. For such situations, the present compositions may be used as anti-solvents for the precipitation of such low solubility solutes from solution in another supercritical or near supercritical solvent, such as carbon dioxide. For example, supercritical carbon dioxide is utilized frequently used in the extrusion process of thermoplastic foams, and the present compositions may be used to precipitation certain materials contained therein.

[0236] It is contemplated also that in certain embodiments it may be desirable to utilize the present compositions when in the supercritical or near supercritical state as a blowing agent.

[0237] The present methods and systems also include forming a one component foam, preferably polyurethane foam, containing a blowing agent in accordance with the present invention. In certain preferably embodiments, a portion of the blowing agent is contained in the foam forming agent, preferably by being dissolved in a foam forming agent which is liquid at the pressure within the container, a second portion of the blowing agent is present as a separate gas phase. In such systems, the contained/dissolved blowing agent performs, in large part, to cause the expansion of the foam, and the separate gas phase operates to impart propulsive force to the foam forming agent. Such one component systems are typically and preferably packaged in a container, such as an aerosol type can, and the blowing agent of the present invention thus preferably provides for expansion of the foam and/or the energy to transport the foam/foamable material from the package, and preferably both. In certain embodiments, such systems and methods comprise charging the package with a fully formulated system (preferably isocyanate/polyol system) and incorporating a gaseous blowing agent in accordance with the present invention into the package, preferably an aerosol type can.

[0238] Any of the methods well known in the art, such as those described in

“Polyurethanes/Chemistry and Technology,” Volumes I and II, Saunders and Frisch, 1962, John Wiley and Sons, New York, NY, which is incorporated herein by reference, may be used or adapted for use in accordance with the foam forming embodiments of the present invention.

[0239] It is contemplated also that in certain embodiments it may be desirable to utilize the present compositions when in the supercritical or near supercritical state as a blowing agent.

8. The Foams

[0240] The invention also relates to all foams, (including but not limited to closed cell foam, open cell foam, rigid foam, flexible foam, integral skin and the like) prepared from a polymer foam formulation containing a blowing agent comprising the compositions of the invention. Applicants have found that one advantage of the foams, and particularly thermoset foams such as polyurethane foams, in accordance with the present invention is the ability to achieve, preferably in connection with thermoset foam embodiments, exceptional thermal performance, such as can be measured by the K-factor or λ , particularly and preferably under low temperature conditions. Although it is contemplated that the present foams, particularly thermoset foams of the present invention, may be used in a wide variety of applications, in certain preferred embodiments the present invention comprises appliance foams in accordance with the present invention, including refrigerator foams, freezer foams, refrigerator/freezer foams, panel foams, and other cold or cryogenic manufacturing applications.

[0241] The foams in accordance with the present invention, in certain preferred embodiments, provide one or more exceptional features, characteristics and/or properties, including: thermal insulation efficiency (particularly for thermoset foams), dimensional stability, compressive strength, aging of thermal insulation properties, all in addition to the low ozone depletion potential and low global warming potential associated with many of the preferred blowing agents of the present invention. In certain highly preferred embodiments, the present invention provides thermoset foam, including such foam formed into foam articles, which exhibit improved thermal conductivity relative to foams made using the same blowing agent (or a commonly used blowing agent HFC-245 fa) in the same amount but without the compound of Formula I in accordance with the present invention. In certain highly preferred embodiments, the thermoset foams, and preferably polyurethane foams, of the present invention exhibit a K-factor (BTU in/hr ft.^{sup.}2° F.) at 40° F. of not greater than about 0.14, more preferably not greater than 0.135, and even more preferably not greater than 0.13. Furthermore, in certain embodiments, it is preferred that the thermoset foams, and preferably the polyurethane foams of the present invention exhibit a K-factor (BTU in/hr ft.^{sup.}2° F.) at 75° F. of not greater than about 0.16, more preferably not greater than 0.15, and even more preferably not greater than 0.145.

[0242] In other preferred embodiments, the present foams exhibit improved mechanical properties relative to foams produced with blowing agents outside the scope of the present invention. For example, certain preferred embodiments of the present invention provide foams and foam articles having a compressive strength which is superior to, and preferably at least about 10 relative percent, and even more preferably at least about 15 relative percent greater than a foam produced under substantially identical conditions by utilizing a blowing agent consisting of cyclopentane. Furthermore, it is preferred in certain embodiments that the foams produced in accordance with the present invention have compressive strengths that are on a commercial basis comparable to the compressive strength produced by making a foam under substantially the same conditions except wherein the blowing agent consists of HFC-245fa. In certain preferred embodiments, the foams of the present invention exhibit a compressive strength of at least about 12.5% yield (in the parallel and perpendicular directions), and even more preferably at least about 13% yield in each of said directions.

EXAMPLES

[0243] The following examples are provided for the purpose of illustrating the present invention but without limiting the scope thereof.

Example 1

[0244] The coefficient of performance (COP) is a universally accepted measure of refrigerant performance, especially useful in representing the relative thermodynamic efficiency of a refrigerant in a specific heating or cooling cycle involving evaporation or condensation of the refrigerant. In refrigeration engineering, this term expresses the ratio of useful refrigeration to the energy applied by the compressor in compressing the vapor. The capacity of a refrigerant represents the amount of cooling or heating it provides and provides some measure of the capability of a compressor to pump quantities of heat for a given volumetric flow rate of refrigerant. In other words, given a specific compressor, a refrigerant with a higher capacity will deliver more cooling or heating power. One means for estimating COP of a refrigerant at specific operating conditions is from the thermodynamic properties of the refrigerant using standard refrigeration cycle analysis techniques (see for example, R. C. Downing, FLUOROCARBON REFRIGERANTS HANDBOOK, Chapter 3, Prentice-Hall, 1988).

[0245] A refrigeration/air conditioning cycle system is provided where the condenser temperature is about 150° F. and the evaporator temperature is about -35° F. under nominally isentropic compression with a compressor inlet temperature of about 50° F. COP is determined for a composition consisting essentially of the compound identified in Table 1 below, over a range of condenser and evaporator temperatures, and each is found to have workable values of COP, capacity and discharge temperature.

TABLE-US-00002 TABLE 1 REFRIGERANT Relative Relative DISCHARGE COMPOSITION
COP CAPACITY TEMPERATURE 1233zdE Workable workable workable 1233zdZ Workable
workable workable 1233ydE Workable workable workable 1233ydZ Workable workable workable
1233zbE Workable workable workable 1233zbZ Workable workable workable 1233xeE Workable
workable workable 1233xeZ Workable workable workable 1233xc Workable workable workable
1233yeE Workable workable workable 1233yc Workable workable workable 1233xf Workable
workable workable

[0246] This example shows that certain of the preferred compounds for use with the present compositions each have a workable energy efficiency and the compressor using the present refrigerant compositions will produce workable discharge temperatures.

Example 2

[0247] The miscibility of a refrigerant composition comprising each of the compounds identified in Table 1 above with various refrigeration lubricants is tested. The lubricants tested are mineral oil (C3), alkyl benzene (Zerol 150), ester oil (Mobil EAL 22 cc and Solest 120), polyalkylene glycol (PAG) oil (Goodwrench Refrigeration Oil for 134a systems), and a poly(alpha-olefin) oil (CP-6005-100). For each refrigerant/oil combination, three compositions are tested, namely 5, 20 and 50 weight percent of lubricant, with the balance of each being the compound of the present invention being tested.

[0248] The lubricant compositions are placed in heavy-walled glass tubes. The tubes are evacuated, the refrigerant compound in accordance with the present invention is added, and the tubes are then sealed. The tubes are then put into an air bath environmental chamber, the temperature of which is varied from about -50° C. to 70° C. At roughly 10° C. intervals, visual observations of the tube contents are made for the existence of one or more liquid phases. The mixtures are found to have acceptable levels of miscibility.

Example 3—Polyol Foam

[0249] This example illustrates the use of blowing agents in accordance with preferred embodiments of the present invention, namely the use of each of the compounds identified in Table 1 above and the use of same for the production of polyol foams in accordance with the present invention. The components of a polyol foam formulation are prepared in accordance with the following Table 2:

TABLE-US-00003 TABLE 2 PBW Polyol Component Voranol 490 50 Vorano0l 391 50 Water 0.5

B-8462 (surfactant) 2.0 Polycat 8 0.3 Polycat 41 3.0 HFO-1234ze 35 Total 140.0 Isocyanate M-20S 123.8 Index 1.10 *Voranol 490 is a sucrose-based polyol and Voranol 391 is a toluene diamine based polyol, and each are from Dow Chemical. B-8462 is a surfactant available from Degussa-Goldschmidt. Polycat catalysts are tertiary amine based and are available from Air Products. Isocyanate M-20S is a product of Bayer LLC.

[0250] The foam is prepared by first mixing the ingredients thereof, but without the addition of blowing agent. Two Fisher-Porter tubes are each filled with about 52.6 grams of the polyol mixture (without blowing agent) and sealed and placed in a refrigerator to cool and form a slight vacuum. Using gas burets, about 17.4 grams of each of the HFCO-1233 compounds in Table 1 are added to each tube, and the tubes are then placed in an ultrasound bath in warm water and allowed to sit for 30 minutes. The isocyanate mixture, about 87.9 grams, is placed into a metal container and placed in a refrigerator and allowed to cool to about 50° F. The polyol tubes were then opened and weighed into a metal mixing container (about 100 grams of polyol blend are used). The isocyanate from the cooled metal container is then immediately poured into the polyol and mixed with an air mixer with double propellers at 3000 RPM's for 10 seconds. The blend immediately begins to froth with the agitation and is then poured into an 8×8×4 inch box and allowed to foam. The foam is then allowed to cure for two days at room temperature. The foam is then cut to samples suitable for measuring physical properties and is found to have acceptable density and K factor.

Example 4—Polystyrene Foam

[0251] This example illustrates the use of blowing agent in accordance with two preferred embodiments of the present invention, namely the use each of the HFCO-1233 compounds described herein as a blowing agent for the production of polystyrene foam. A testing apparatus and protocol has been established as an aid to determining whether a specific blowing agent and polymer are capable of producing a foam and the quality of the foam. Ground polymer (Dow Polystyrene 685D) and blowing agent consisting essentially of each of the HFCO-1233 compounds described herein are combined in a vessel. The vessel volume is 200 cm.^{sup.3} and it is made from two pipe flanges and a section of 2-inch diameter schedule 40 stainless steel pipe 4 inches long. The vessel is placed in an oven, with temperature set at from about 190° F. to about 285° F., preferably for polystyrene at 265° F., and remains there until temperature equilibrium is reached.

[0252] The pressure in the vessel is then released, quickly producing a foamed polymer. The blowing agent plasticizes the polymer as it dissolves into it. The resulting density of the two foams thus produced using this method is determined and found to be acceptable.

Example 5A—Polystyrene Foam

[0253] This example demonstrates the performance of each of the HFCO-1233 compounds described herein alone as a blowing agent for polystyrene foam formed in a twin screw type extruder. The apparatus employed in this example is a Leistritz twin screw extruder having the following characteristics: [0254] 30 mm co-rotating screws [0255] L:D Ratio=40:1

The extruder is divided into 10 sections, each representing a L:D of 4:1. The polystyrene resin was introduced into the first section, the blowing agent was introduced into the sixth section, with the extrudate exiting the tenth section. The extruder operated primarily as a melt/mixing extruder. A subsequent cooling extruder is connected in tandem, for which the design characteristics were:

[0256] Leistritz twin screw extruder [0257] 40 mm co-rotating screws [0258] L:D Ratio=40:1 [0259] Die: 5.0 mm circular

Polystyrene resin, namely Nova Chemical—general extrusion grade polystyrene, identified as Nova 1600, is feed to the extruder under the conditions indicated above. The resin has a recommended melt temperature of 375° F.-525° F. The pressure of the extruder at the die is about 1320 pounds per square inch (psi), and the temperature at the die is about 115° C.

[0260] A blowing agent consisting essentially of each of the HFCO-1233 compounds described herein alone is added to the extruder at the location indicated above, with about 0.5% by weight of talc being included, on the basis of the total blowing agent, as a nucleating agent. Foam is produced

using the blowing agent at concentrations of 10% by weight, 12% by weight, and 14% by weight, in accordance with the present invention. The density of the foam produced is in the range of about 0.1 grams per cubic centimeter to 0.07 grams per cubic centimeter, with a cell size of about 49 to about 68 microns. The foams, of approximately 30 millimeters diameter, are visually of very good quality, very fine cell size, with no visible or apparent blow holes or voids.

Example 5B—Polystyrene Foam

[0261] This procedure of Example 5C is repeated except that the foaming agent comprises about 50% by weight of each of the HFCO-1233 compounds described herein and 50% by weight of HFC-245fa and nucleating agent in the concentration indicated in Example 5. Foamed polystyrene is prepared at blowing agent concentrations of approximately 10% and 12%. The density of the foam produced is about 0.09 grams per cubic centimeter, with a cell size of about 200 microns. The foams, of approximately 30 millimeters diameter, are visually of very good quality, fine cell structure, with no visible or apparent voids.

Example 5C—Polystyrene Foam

[0262] This procedure of Example 5 is repeated except that the foaming agent comprises about 80% by each of the HFCO-1233 compounds described herein and 20% by weight of HFC-245fa and nucleating agent in the concentration indicated in Example 5. Foamed polystyrene is prepared at blowing agent concentrations of approximately 10% and 12%. The density of the foam produced is about 0.08 grams per cubic centimeter, with a cell size of about 120 microns. The foams, of approximately 30 millimeters diameter, are visually of very good quality, fine cell structure, with no visible or apparent voids.

Example 5D—Polystyrene Foam

[0263] This procedure of Example 5 is repeated each of the HFCO-1233 compounds described herein alone except that the nucleating agent is omitted. The foams' density was in the range of 0.1 grams per cubic centimeter, and the cell size diameter is about 400. The foams, of approximately 30 millimeters diameter, are visually of very good quality, fine cell structure, with no visible or apparent voids.

Example 6—Polyurethane Foam

[0264] This example demonstrates the performance of each of the HFCO-1233 compounds described herein, used in combination with hydrocarbon co-blowing agents, and in particular the utility of compositions comprising each of the HFCO-1233 compounds described herein alone and cyclopentane co-blowing agents to produce polyurethane foams having acceptable compressive strength performance.

[0265] A commercially available, refrigeration appliance-type polyurethane foam formulation (foam forming agent) is provided. The polyol blend consisted of commercial polyol(s), catalyst(s), and surfactant(s). This formulation is adapted for use in connection with a gaseous blowing agent. Standard commercial polyurethane processing equipment is used for the foam forming process. A gaseous blowing agent combination was formed comprising each of the HFCO-1233 compounds described herein in a concentration of approximately 60 mole percent, and cyclopentane in a concentration of approximately 40 mole percent of the total blowing agent. This example illustrates acceptable physical property performance, including compressive strength and K-factor performance of combinations of each of the HFCO-1233 compounds described herein in combination with cyclopentane co-blowing agent.

Example 7—Polyurethane Foam K-Factors

[0266] This example demonstrates the performance of blowing agents comprising each of the HFCO-1233 compounds described herein in combination with each of the HFC co-blowing agents mentioned above in connection with the preparation of polyurethane foams. The same foam formulation, equipment and procedures used in Examples 5 and 6 are used, with the exception of the blowing agent. A blowing agent is prepared comprising each of the HFCO-1233 compounds described herein in a concentration of approximately 80 weight percent of the total blowing agent,

and each of the HFC co-blowing agents mentioned above in a concentration of approximately 20 weight percent of the total blowing agent. Foams are then formed using this blowing agent and the k-factors of the foam are measured and found to be acceptable.

Example 8—Polyurethane Foam K-Factors

[0267] A further experiment is performed using the same polyol formulation and isocyanate as in Examples 5 and 6. The foam is prepared by hand mix. The blowing agent consists of a compound in accordance with each of the HFCO-1233 compounds described herein in about the same mole percentage of the foamable composition as the blowing agent in Examples 5 and 6. Acceptable foams are formed.

Example 9—Polyurethane Foam K-Factors

[0268] A further experiment is performed using the same polyol formulation and isocyanate as in Examples 5 and 6. The foam is prepared by hand mix. A series of blowing agent consisting of a combination of each of the HFCO-1233 compounds described herein and each of methanol, propanol, isopropanol, butanol, isobutanol and t-butanol in a 50:50 mole ratio, each combination being present in the blowing agent composition in about the same mole percentage of the foamable composition as the blowing agent in Examples 5 and 6. In each case an acceptable foam is formed.

Example 10—Polyurethane Foam K-Factors

[0269] A further experiment is performed using the same polyol formulation and isocyanate as in Examples 5 and 6. The foam is prepared by hand mix. A series of blowing agent consisting of a combination of each of the HFCO-1233 compounds described herein and each of and each of the following additional compounds: iso-pentane, normal-pentane and cyclo-pentane. Three blowing agents are formed in combination with each additional compound in HFCO-1233:additional compound mole ratios of 25:75, 50:50, and 75:25. Each blowing agent composition is present in about the same mole percentage of the foamable composition as the blowing agent in Examples 5 and 6. An acceptable foam is formed in each case.

Example 11—Polyurethane Foam K-Factors

[0270] A further experiment is performed using the same polyol formulation and isocyanate as in Examples 5 and 6. The foam is prepared by hand mix. A series of blowing agents consisting of a combination of each of the HFCO-1233 compounds described herein and each of the following additional compounds: water and CO₂. Three blowing agents are formed in combination with each additional compound in HFCO-1233:additional compound mole ratios of 25:75, 50:50, and 75:25. Each blowing agent composition is present in about the same mole percentage of the foamable composition as the blowing agent in Examples 5 and 6. An acceptable foam is formed in each case.

Example 12—Polyurethane Foam K-Factors

[0271] A further experiment is performed using the same polyol formulation and isocyanate as in Examples 5 and 6. The foam is prepared by hand mix. A series of blowing agent consisting of a combination of each of the HFCO-1233 compounds described herein and each of HFO-1234ye-trans(E) (having a boiling point of 15 C) and HFO-1234ye-cis(Z) (having a boiling point of 24 C), in combination with each HFCO-1233 in a 50:50 mole ratio, each combination being present in the blowing agent composition in about the same mole percentage of the foamable composition as the blowing agent in Examples 5 and 6. An acceptable foam is formed in each case.

Example 13—Polyurethane Foam K-Factors

[0272] A further experiment is performed using the same polyol formulation and isocyanate as in Examples 5 and 6. The foam is prepared by hand mix. A blowing agent consisting of a combination of each of the HFCO-1233 compounds described herein and trans-1,2 dichloroethylene, in an HFCO-1233:trans-1,2 dichloroethylene mole ratio of 75:25, with the blowing agent composition being in about the same mole percentage of the foamable composition as the blowing agent in Examples 5 and 6. An acceptable foam is formed.

Example 14—Polyurethane Foam K-Factors

[0273] A further experiment is performed using the same polyol formulation and isocyanate as in

Example 9. The foam is prepared by hand mix. The blowing agent consisting of a combination of each of the HFCO-1233 compounds described herein and methyl formate, in a 75:25 mole ratio, the combination being present in the blowing agent composition in about the same mole percentage of the foamable composition as the blowing agent in Examples 5 and 6. An acceptable foam is formed in each case.

Example 15—Silicon Solvent

[0274] A series of compositions were prepared with each composition consisting of each of the HFCO-1233 compounds described herein. Each composition is transferred to a glass container. A silicon lubricant, particularly a high-viscosity (12,500 cP) silicone oil, was added to the composition to a concentration of about 10 weight percent. This resulted in a homogeneous, single-phase solution, demonstrating that each of the HFCO-1233 compounds dissolves silicone based lubricant oils.

Example 16—HFCO-1233/Trans-1,2-dichloroethylene

[0275] A series of compositions were prepared with each composition consisting of each of the HFCO-1233 compounds described herein and trans-1,2-dichloroethylene in HFCO-1233:trans-1,2-dichloroethylene weight ratios of 25:75 and 50:50. Each combination is then added to a glass container. A silicon lubricant, particularly a high-viscosity (12,500 cP) silicone oil is added to each solvent to a concentration of about 10 weight percent. This results in a homogeneous, single-phase solution, demonstrating that this combination dissolves silicone oil.

Example 17—Cleaning Agent

[0276] A metal coupon was coated with rosin-based solder flux and allowed to dry. The coupon was weighed and then dipped in series of compositions consisting of each of the HFCO-1233 compounds described herein. The coupon was removed, allowed to dry and reweighed to determine how much solder flux was removed. In duplicate runs, an average of 25% by weight of the flux was removed.

Example 18—HFCO-1233/Methanol as Cleaning Agent

[0277] A metal coupon is coated with rosin-based solder flux and allowed to dry. The coupon is weighed and then dipped in series of compositions consisting of each of the HFCO-1233 compounds described herein and methanol in several different concentrations ranging from about 1% to about 10% (and even more preferably from about 1% to about 5%), including about 1%, about 2%, about 3%, about 5% and about 10% by weight. The coupon is removed, allowed to dry and reweighed to determine how much solder flux is removed. In duplicate runs, flux is removed.

Example 19—Extractant

[0278] A medicament, particularly a plant-derived Artemisinin which is an anti-malarial drug, is extracted from the *Artemisia annua* plant. A sample of Artemisinin was weighed into a vial. A series of compositions consisting of each of the HFCO-1233 compounds described herein was added to the vial until the Artemisinin dissolved. The results showed that medicaments, particularly plant-derived medicaments such as Artemisinin is soluble in each of the HFCO-1233 compounds described herein, demonstrating that such compounds can be used to extract the drug from biomass.

Example 20—Solvent—Mineral Oil

[0279] A hydrocarbon lubricant, specifically mineral oil, was added to vials containing, respectively a series of compositions consisting of each of the HFCO-1233 compounds described herein and methanol in an approximate 98:2 weight ratio, in an approximate 96:4 weight ratio and a HFCO-1233/methanol/pentane in an approximate 92:2:6 weight ratio. In all cases homogeneous, single-phase solutions are formed at concentrations greater than 10% by weight of the mineral oil.

Example 21—Aerosol

[0280] A sprayable aerosol was prepared by adding a series of compositions consisting of each of the HFCO-1233 compounds described herein to an aerosol can, sealing the can by crimping an aerosol valve in place and adding HFC-134a propellant to a concentration of about 14% by weight of the 134a and about 76% by weight of the HFCO-1233. Hydraulic fluid was applied to a metal

coupon with a cotton swab and the coupon was weighed. Each of the HFCO-1233-containing aerosol was sprayed onto the metal substrate for 10 seconds. The coupon was allowed to dry and was reweighed. Approximately 60% by weight of the hydraulic fluid was removed.

Example 22—Solvent—PAG

[0281] A synthetic lubricant, specifically polyalkyleneglycol (PAG) lubricant, and more specifically a PAG consisting essentially of 2 or more oxypropylene groups and having a viscosity of from about 10 to about 200 centistokes at about 37° C. (sold under the trade designation ND-8 by Idemitsu Kosan) are added to a vial containing a series of compositions consisting of each of the HFCO-1233 compounds described herein. A homogeneous, single-phase solution is formed at concentrations greater than 10% by weight of the PAG. The properties of the synthetic lubricant ND-8 are identified below in the Table below.

TABLE-US-00004 ND-8 PROPERTIES Viscosity, @ EO:PO Molecular Property 40° C. cSt Ratio Weight* 42.3 0:1 930 *molecular weight is Number Average Molecular Weight

Example 23—HFCO-1233 and Cosolvents

[0282] The PAG lubricant described in Example 22 above is added to vials containing, respectively, each of the HFCO-1233 compounds mentioned above in combination with: (a) methanol in an approximate 98:2 HFCO:methanol weight ratio; (b) pentane in an approximate 96:4 HFCO:pentane weight ratio; and (c) methanol/pentane in an approximate 92:2:6 HFCO:methanol:pentane weight ratio. In all cases homogeneous, single-phase solutions are formed at concentrations greater than 10% by weight of the PAG oil.

Example 24

[0283] This example illustrates the performance of one embodiment of the present invention in which a refrigerant composition comprises each of the HFCO-1233 compounds described above wherein a large proportion, and preferably at least about 75% by weight and even more preferably at least about 90% by weight, of the refrigerant composition is each of said HFCO-1233 compounds. More particularly, this example is illustrative of such a composition being used as a working fluid in a refrigerant system, High Temperature Heat Pump and Organic Rankine Cycle system. An example of the first system is one having an Evaporation Temperature of about of 35° F. and a Condensing Temperature of about 150° F. For the purposes of convenience, such heat transfer systems, that is, systems having an evaporator temperature of from about 35° F. to about 50° F. and a CT of from about 80° F. to about 120° F., are referred to herein as “chiller” or “chiller AC” systems. The operation of each of such systems is found to be acceptable using R-123 for the purposes of comparison.

Example 25

[0284] This example illustrates the performance of one embodiment of the present invention in which a refrigerant composition comprises each of the HFCO-1233 compounds mentioned above wherein a large proportion, and preferably at least about 75% by weight and even more preferably at least about 90% by weight, of the composition comprises each of the HFCO-1233 compounds mentioned above. More particularly, such a composition is used as a replacement for HFC-134a in four refrigerant systems. The first system is one have an evaporator temperature (ET) of about 20° F. and condenser temperature (CT) of about 130° F. For the purposes of convenience, such heat transfer systems, that is, systems having an ET of from about 0 to about 35 and a CT of from about 80° F. to about 130° F., are referred to herein as “medium temperature” systems. The second system is one have an ET of about −10° F. and a CT of about 110° F. For the purposes of convenience, such heat transfer systems, that is, systems having an evaporator temperature of from about −20° F. to about 20° F. and a CT of from about 80° F. to about 130° F., are referred to herein as “refrig/freezer” systems. The third system is one have an ET of about of 35° F. and a CT of about 150° F. For the purposes of convenience, such heat transfer systems, that is, systems having an evaporator temperature of from about 30° F. to about 60° F. and a CT of from about 90° F. to about 200° F., are referred to herein as “automotive AC” systems. The fourth system is one have an

ET of about 40° F. and a CT of about 60° F. For the purposes of convenience, such heat transfer systems, that is, systems having an evaporator temperature of from about 35° F. to about 50° F. and a CT of from about 80° F. to about 120° F., are referred to herein as “chiller” or “chiller AC” systems. The operation of each of such systems using each of the compositions, in comparison to R-134a, is found to be acceptable.

[0285] Based on the above examples, many of the important refrigeration system performance parameters are relatively close to the parameters for many previously used refrigerants, such as R-134a. Since many existing refrigeration systems have been designed for these refrigerants, including R-134a or for other refrigerants, those skilled in the art will appreciate the substantial advantage of a low GWP and/or a low ozone depleting refrigerant that can be used as replacement for R-134a or like refrigerants with relatively minimal modifications to the system. It is contemplated that in certain embodiments the present invention provided retrofitting methods which comprise replacing the refrigerant in an existing system with a composition of the present invention, preferably a composition comprising at least about 90% by weight and/or consists essentially of the HFCO-1233 compounds mentioned above, without substantial modification of the system. In certain preferred embodiments the replacement step is a drop-in replacement in the sense that no substantial redesign of the system is required and no major item of equipment needs to be replaced in order to accommodate the refrigerant of the present invention.

Claims

1. A solvent extraction method comprising solvent extracting a material by contacting said material with at least one monochlorotrifluoropropene selected from the group consisting of transCF₃CH=CClH (1233zdE), cisCF₃CH=CClH (1233zdZ), transCHF₂CF=CClH (1233ydE), cisCHF₂CF=CClH (1233ydZ), transCHF₂CH=CClF (1233zbE), cisCHF₂CH=CClF (1233zbZ), transCHF₂CCl=CHF (1233xeE), cisCHF₂CCl=CHF (1233xeZ), CH₂FCCl=CF₂ (1233xc), transCHFC1CF=CFH (1233yeE), cisCHFC1CF=CFH (1233yeZ), CH₂C1CF=CF₂ (1233yc), CF₂C1CF=CH₂ (1233xf), and combinations of two or more of these.
 2. The method of claim 1 wherein said at least one monochlorotrifluoropropene is selected from the group consisting of transCF₃CH=CClH (1233zdE), cisCF₃CH=CClH (1233zdZ), transCHF₂CF=CClH (1233ydE), cisCHF₂CF=CClH (1233ydZ) and combinations of two or more of these.
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